

Mean Field Theory

Markov models for image labelling,
the weak membrane and binocular stereo
Notes for Probabilistic Models of the Visual Cortex

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What this lecture covers

Two Markov field models

Mean field free energies

Minimising the free energy

A cooperative stereo model

Neural implementation

Summary

From semantic segmentation to other Markov models

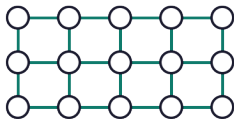
The segmentation lecture built one machine. A Markov random field (MRF) puts a variable at every pixel and joins neighbouring pixels by edges. An energy scores every labelling: a unary term at each node says what the image says at that pixel, and a pairwise term on each edge says whether two neighbouring labels are compatible. The energy defines a probability distribution, and mean field theory (MFT) replaces that coupled distribution by a factorised one, fitted by minimising a free energy.

The same machine handles other visual tasks. Three are studied here: labelling the pixels of an image, finding depth from the images in the two eyes, and smoothing a noisy image while keeping its sharp boundaries. The new question is: how does the same machinery apply to stereo and to discontinuities, and how could neurons carry it out?

The notation follows the chapter on mean field theory (Yuille, 2011): the image I is now the input z , and the factorised Q with beliefs q_i is now B with pseudo-marginals b_i .

From semantic segmentation to other Markov models

semantic segmentation



an energy on a graph:

unary evidence at the nodes

pairwise context on the edges



image labelling



z

x

intensities
to labels

binocular stereo



z^L

z^R

two images
to a disparity

weak membrane



smooth,
with breaks

Figure 1: The model of the segmentation lecture and the three tasks studied here. Inputs are grey, outputs blue. Orange marks what is new: the shift between the two images of a stereo pair, and the break in a smooth output.

Two vision models: the MRF and the weak membrane

Both models live on the image lattice $\mathcal{D}$, the regular grid of pixel positions. At every pixel $i \in \mathcal{D}$ there is an observed value z_i , the input, and an unknown value x_i , the output, which has to be inferred:

$$z = \{z_i : i \in \mathcal{D}\} \text{ input,} \quad x = \{x_i : i \in \mathcal{D}\} \text{ output.}$$

A variable that belongs to the model and is never observed is called hidden. In the Markov random field with input z and output x , each pixel has an observed node z_i and a hidden node x_i .

The weak membrane model smooths a noisy image and keeps its sharp boundaries. It has additional hidden variables $y = \{y_{ij}\}$, the line processes. A line process y_{ij} is a binary variable that sits on the edge between two neighbouring pixels i and j . It records whether the output is allowed to jump between the two pixels: $y_{ij} = 0$ keeps x_i and x_j tied together, and $y_{ij} = 1$ cuts the tie.

Two vision models: the MRF and the weak membrane

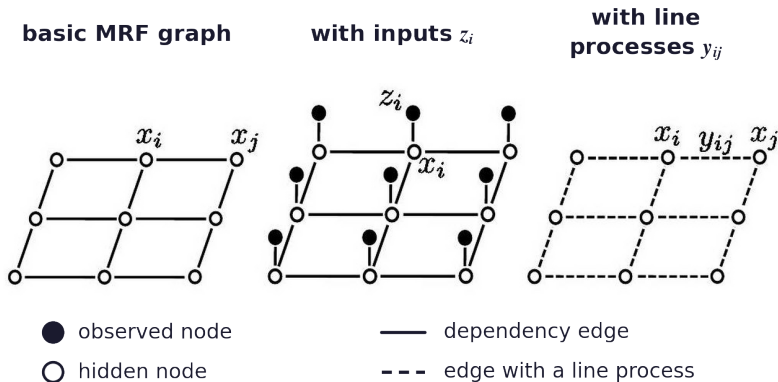


Figure 2: Graphs for the two models (Yuille, 2011). The lattice is drawn in perspective. An open circle is a hidden node x_i , a filled circle an observed node z_i , and a solid line a dependency between the two variables it joins. The first two graphs are the MRF, without and with its inputs. The third is the weak membrane, its inputs left out: each dashed edge stands for a line process y_{ij} between x_i and x_j .

Image labelling: intensity values in, a label at every pixel

The first application of the MRF is image labelling. The input is a grey-level image and the output gives every pixel a label from a finite set $\mathcal{L}$:

$$z = \{z_i : i \in \mathcal{D}\}, \quad z_i \in \{0, \dots, 255\}; \quad x = \{x_i : i \in \mathcal{D}\}, \quad x_i \in \mathcal{L}.$$

The input z_i is the intensity at pixel i , an integer from 0 for black to 255 for white. The size of the label set depends on the task. Edge detection has two labels, edge and non-edge. The Microsoft Research Cambridge (MSRC) data set, a standard benchmark for labelling photographs, has 23 labels such as sky, grass and cow.

A label can be estimated at each pixel from the intensities around that pixel alone. Labels estimated pixel by pixel are noisy, because a small window is often ambiguous. The MRF adds costs on the edges between neighbouring pixels, the pairwise potentials, which supply the spatial context that resolves the ambiguity.

Image labelling: intensity values in, a label at every pixel

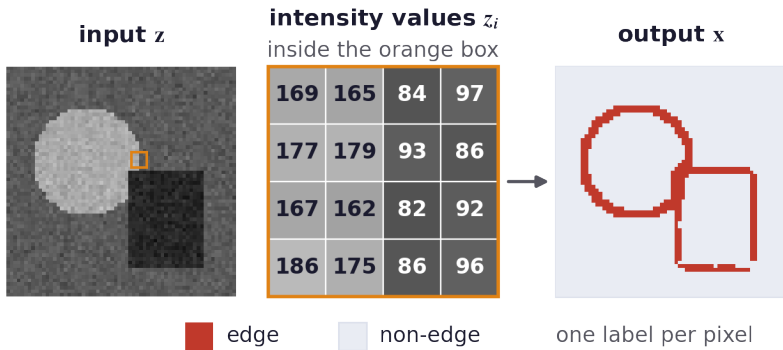


Figure 3: An input of 64×64 pixels with a bright disk and a dark rectangle on a grey ground, plus noise. The orange box marks a block of 4×4 pixels on the rim of the disk, and the middle panel prints the 16 intensity values z_i inside it: about 170 on the disk and about 90 on the ground. The output colours each pixel by its label: red for edge, pale grey for non-edge.

Binocular stereo: two images in, a disparity at every pixel

The second application is binocular stereo. The input is a pair of images of the same scene, from a left and a right camera or from the two eyes, and the output is a disparity at every pixel:

$z = (z^L, z^R)$ left and right images, $x_i = \text{disparity at pixel } i$.

Each camera has an optical centre, the point through which all its light rays pass: O_L and O_R . A point P of the scene is seen where its ray to the optical centre crosses the image plane, at P_L and at P_R . The plane through P , O_L and O_R cuts each image plane in a line, the epipolar line: in the figure, the solid lines ending at the dots E_L and E_R on the dashed baseline from O_L to O_R . The match of a point of the left epipolar line always lies in the right epipolar line, so the search runs along one line.

The disparity is the displacement between the positions of P_L and P_R along their epipolar lines. Points P_1 and P_2 on the same ray as P share the image point P_L , and their dotted rays to O_R cross the right epipolar line at different places: the disparity encodes the depth, which is roughly inversely proportional to its magnitude.

Binocular stereo: two images in, a disparity at every pixel

the geometry of stereo

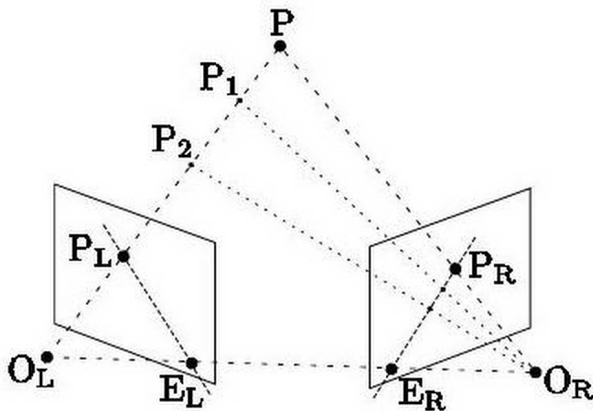


Figure 4: The geometry of stereo (Yuille, 2011). The two quadrilaterals are the image planes. P_2 , P_1 and P lie on one ray from O_L , share the image point P_L , and project to three different places on the right epipolar line.

Binocular stereo: two images in, a disparity at every pixel

The graph of the stereo MRF has one node per pixel, holding the disparity x_i , a signed number of pixels: for cameras that converge on a point of the scene it is negative for nearer points and positive for points beyond. Current models put a weak smoothness prior on the disparity, the assumption that neighbouring pixels have similar disparities, over the whole lattice: a node is joined to its neighbours along its own epipolar line and to the neighbours on the lines above and below.

Earlier models assumed that the disparities on different epipolar lines are independent. The lattice then falls apart into separate chains, and on a chain exact inference is possible by dynamic programming: one sweep along the chain keeps, for every value of the current node, the lowest cost of everything before it, and a sweep back reads off the best configuration. The price is that neighbouring lines can disagree, which shows as streaks in the depth map.

Binocular stereo: two images in, a disparity at every pixel

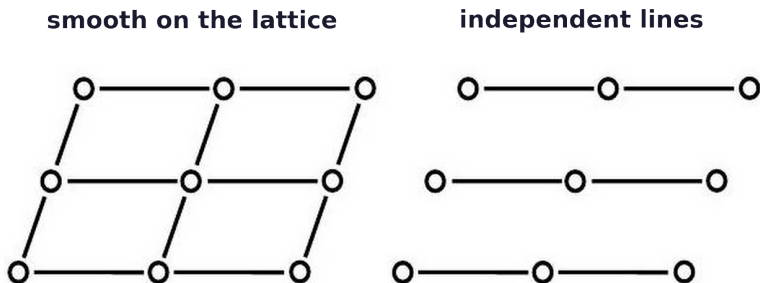


Figure 5: The two graph structures used for stereo (Yuille, 2011). Each row of nodes is one epipolar line. With the vertical edges the rows constrain each other; without them every row is solved on its own.

The MRF is a Gibbs distribution on a graph

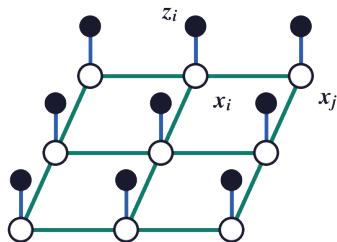
Both applications are the same mathematical object. Take a graph $G = (\mathcal{V}, \mathcal{E})$: the nodes $\mathcal{V}$ are the image pixels $\mathcal{D}$, and the edges $\mathcal{E}$ join neighbouring pixels. Each node i carries a random variable x_i . An energy $E(x, z)$ gives every configuration x of all the nodes a number, and the probability of a configuration falls exponentially with its energy. This is a Gibbs distribution:

$$P(x | z) = \frac{1}{Z(z)} \exp \left\{ - \underbrace{\sum_{i \in \mathcal{V}} \phi_i(x_i, z)}_{\text{unary potentials}} - \underbrace{\sum_{ij \in \mathcal{E}} \psi_{ij}(x_i, x_j)}_{\text{pairwise potentials}} \right\}.$$

The unary potential $\phi_i(x_i, z)$ is the cost of giving node i the value x_i when the input is z . The pairwise potential $\psi_{ij}(x_i, x_j)$ is the cost of the pair of values on the edge ij ; the sum counts every edge once. The constant $Z(z)$, the partition function, is the sum of $\exp\{-E(x, z)\}$ over all configurations, so that the probabilities add up to one.

The MRF is a Gibbs distribution on a graph

$P(x|z)$ is a posterior distribution, conditioned on the input z . A Markov random field whose potentials depend on the input in this way is called a conditional random field (CRF).



unary potential $\phi_i(x_i, z)$

how well the label
fits the input

pairwise potential $\psi_{ij}(x_i, x_j)$

how compatible two
neighbouring labels are

one term for every edge ij in $\mathcal{E}$

Figure 6: A 3×3 piece of the graph. Open circles are the hidden nodes x_i and filled circles the observed inputs z_i . Every blue link stands for one unary potential, which ties a label to the input. Every teal link stands for one pairwise potential, which ties the labels of two neighbouring pixels to each other.

The potentials for image labelling and for stereo

The two applications differ only in their potentials:

$$\text{labelling: } \phi_i(x_i, z) = g_{x_i}(z)_i,$$

$$\text{stereo: } \phi(x_i, z^L, z^R) = |f(z^L)_i - f(z^R)_{i+x_i}|.$$

A filter is a function computed from a small window of the image. For labelling, g is a non-linear filter applied to a window around pixel i . It has one output for every label, $g_{x_i}(z)_i$ being the cost of label x_i at pixel i . An example is AdaBoost, a classifier that combines many simple tests on the window and is trained on labelled images.

For stereo, f is a vector-valued filter: $f(z^L)_i$ is a short list of numbers that describes the left image around pixel i . A disparity x_i claims that pixel i of the left image matches pixel $i + x_i$ of the right image, and $|\cdot|$ is the L_1 norm, the sum of the absolute differences of the two lists. Similar filter responses give a small potential, so a good match is cheap.

The pairwise potentials impose a smoothness prior: neighbouring pixels tend to have similar labels or similar disparities.

The potentials for image labelling and for stereo

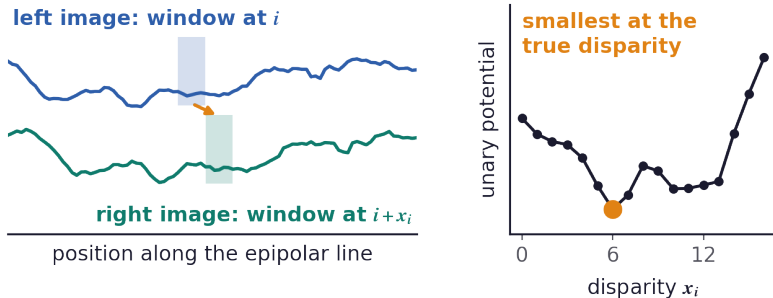


Figure 7: The stereo unary potential on one epipolar line. The blue and the teal curve are the intensities of the left and the right image along the line; the right image is the left one shifted by 6 pixels, plus noise. The shaded bands are the window at pixel i and the window at $i + x_i$, and the orange arrow is the disparity x_i . The second plot gives the potential for every candidate disparity from 0 to 16: the smallest value is at the true disparity 6, and a second, shallower dip between 10 and 13 is the kind of rival that the smoothness prior removes.

The goal: the MAP estimate and the marginals

Two properties of the distribution are wanted:

$$x^* = \arg \max_x P(x | z) \quad \text{MAP estimate,}$$

$$p_i(x_i) = \sum_{x/i} P(x | z), \quad \forall i \in \mathcal{V} \quad \text{marginals.}$$

The maximum a posteriori (MAP) estimate is the single most probable configuration of all the nodes together. A marginal is the distribution of one node; the symbol x/i means all the nodes except i , so $p_i(x_i)$ adds up the probabilities of every configuration in which node i has the value x_i . The marginal says how certain the model is about one pixel.

With $|\mathcal{L}|$ labels on $|\mathcal{V}|$ nodes there are $|\mathcal{L}|^{|\mathcal{V}|}$ configurations: 2^{4096} for two labels on a 64×64 image. Computed directly, both the maximum and the sums run over exponentially many configurations, so approximate methods are used.

The goal: the MAP estimate and the marginals

A chain of three binary nodes can still be enumerated. Take the energy $\sum_i \phi_i x_i + \sum_{ij} \psi_{ij} x_i x_j$ with $\phi = (-1.2, 0.4, -0.8)$ and $\psi_{12} = \psi_{23} = -1$. The MAP estimate is 111, with probability 0.597, and the marginals $p_i(x_i=1)$ are 0.870, 0.773 and 0.820.

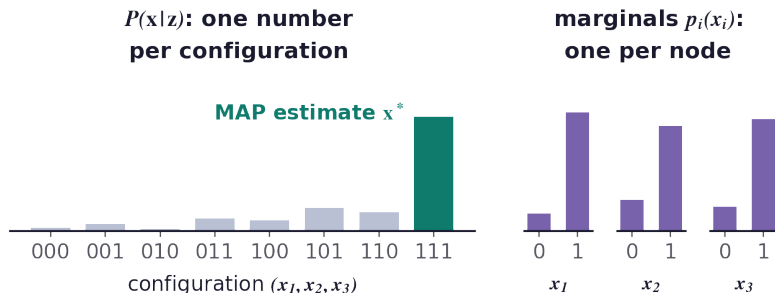


Figure 8: Each bar of the first chart is the probability of one configuration; the tallest, in teal, is the MAP estimate. In each small chart the bar over 1 is $p_i(x_i=1)$, the sum of the four bars of the first chart whose configuration has $x_i = 1$.

The weak membrane model labels the discontinuities

The second model addresses image segmentation, denoising and depth smoothing. The input z holds noisy intensity or depth values, and the output x is a smoothed version of z on the same lattice. Plain smoothing removes the noise and also blurs the boundaries between regions, which are the most informative places of the image.

The name describes the remedy. Picture an elastic membrane, a rubber sheet, pulled towards the data at every pixel. Its elasticity resists differences between neighbouring values, so it follows the data smoothly. The membrane is weak: it can tear. Where the data jump, tearing the membrane costs less than stretching it, and the two sides are then smoothed separately (Blake and Zisserman, 1987).

The weak membrane model labels the discontinuities

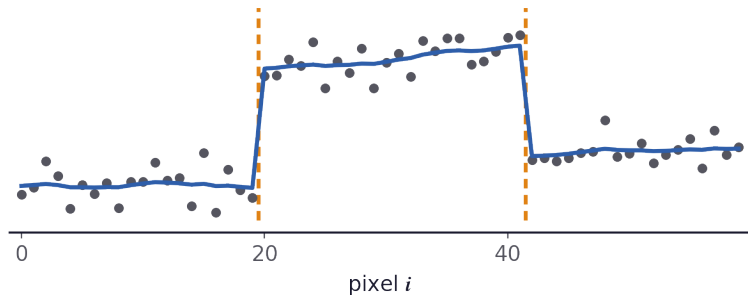
The tears are explicit variables of the model, the line processes of Geman and Geman (1984):

$$y = \{y_{ij} : (i,j) \in \mathcal{E}\}, \quad y_{ij} \in \{0,1\}.$$

There is one hidden variable y_{ij} for every edge of the lattice, and $y_{ij} = 1$ labels a discontinuity between pixels i and j . The word process is used as in random process: the y_{ij} form a random field of their own, living on the edges while x lives on the nodes. In a two-dimensional image the broken edges line up along the boundaries of the regions, which gives the word line.

The model therefore does two things at once. It smooths the image, and it labels the discontinuities, so its output is a segmentation as well as a cleaner image.

The weak membrane model labels the discontinuities



- input z_i : noisy values
- output x_i : smooth in pieces
- - - line process $y_{ij}=1$: a discontinuity between i and j

Figure 9: A one-dimensional example with 60 pixels. The grey dots are the input z_i : three levels, the middle one a slow ramp, with added noise. The blue line is the output x_i . The two orange dashed lines stand between pixels 19 and 20 and between pixels 41 and 42, where the line process is $y_{ij} = 1$. The output is smooth inside each of the three pieces and jumps across the two breaks.

A Gaussian likelihood and a piecewise smooth prior

The weak membrane is a generative model: a prior $P(x, y)$ on the hidden variables and a likelihood $P(z | x)$ for the input,

$$P(z | x) = \prod_{i \in \mathcal{D}} \sqrt{\frac{\tau}{\pi}} \exp \{ -\tau (z_i - x_i)^2 \}, \quad P(x, y) \propto \exp \{ -E(x, y) \}.$$

The likelihood says that z is x plus independent zero-mean Gaussian noise at every pixel; a large τ means little noise. The variables x_i and z_i are continuous and the line processes are binary. The prior has the energy

$$E(x, y) = \underbrace{A \sum_{(i,j) \in \mathcal{E}} (x_i - x_j)^2 (1 - y_{ij})}_{\text{smoothness}} + \underbrace{B \sum_{(i,j) \in \mathcal{E}} y_{ij}}_{\text{a penalty for every break}}$$

With $y_{ij} = 0$ the edge pays $A(x_i - x_j)^2$, which keeps neighbouring values similar. With $y_{ij} = 1$ the factor $(1 - y_{ij})$ switches the smoothness off and the edge pays the fixed price B . The prior is piecewise smoothness: $x_i \approx x_j$ unless $y_{ij} = 1$.

A Gaussian likelihood and a piecewise smooth prior

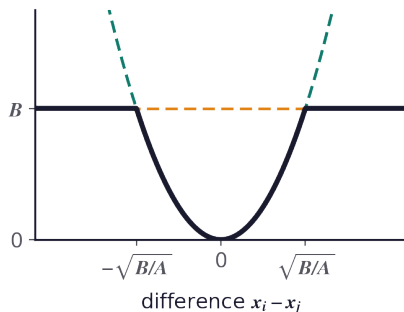
One edge with difference $d = x_i - x_j$ costs Ad^2 if it is kept and B if it is broken. Choosing the cheaper option gives

$$\min_{y_{ij} \in \{0,1\}} \{Ad^2(1 - y_{ij}) + By_{ij}\} = \min(Ad^2, B),$$

and the two costs are equal at $|d| = \sqrt{B/A}$. Differences of the output below this threshold are cheaper to keep, and larger ones are cheaper to break. The example of 60 pixels uses $\tau = 60$, $A = 600$ and $B = 6$, so the threshold is 0.1. A jump in the data has to be larger than that to be kept, because a small jump can be spread over several pixels at little cost: for these constants a break pays once an isolated jump exceeds about 0.25. The jumps of the signal, 0.5 and 0.43, are well above, and the noise, of standard deviation 0.06, is well below.

The line processes can also be summed out in place of being minimised out. One edge then contributes $-\log(e^{-Ad^2} + e^{-B})$, a smooth version of the same curve. The minimum is what a MAP estimate of x and y together uses; the sum is what an estimate of x alone uses.

A Gaussian likelihood and a piecewise smooth prior



$y_{ij} = 0$: **keep the edge**

the cost is $\Lambda(x_i - x_j)^2$

$y_{ij} = 1$: **break the edge**

the cost is B , whatever
the difference

solid line: the cheaper
of the two costs

Figure 10: The cost of one edge as a function of the difference $x_i - x_j$. The dashed teal parabola is the cost of keeping the edge, the dashed orange level the cost of breaking it, and the solid black curve the cheaper of the two. Beyond $\pm\sqrt{B/A}$ the cost stops growing, which is why a large jump survives the smoothing.

Inference in the weak membrane model is hard

By Bayes' rule the posterior of the hidden variables is

$$P(x, y | z) \propto P(z | x) P(x, y) \propto \exp \left\{ -\tau \sum_i (z_i - x_i)^2 - E(x, y) \right\}.$$

The posterior couples the continuous variables x and the discrete variables y : a hybrid problem. For a fixed y the best x solves a set of linear equations, but there are $2^{|\mathcal{E}|}$ settings of y , and a 64×64 lattice has 8064 edges. With y minimised out, the energy of x is a sum of curves $\min(Ad^2, B)$ with many local minima, points that are lower than everything near them and higher than the best.

Better priors make the problem harder still. More advanced models add higher-order coupling terms between neighbouring line processes, such as products $y_{ij} y_{kl}$. In the basic model every break is priced on its own, so noise produces short isolated breaks. A coupling term makes a break cheaper when the break next to it continues it. The coupling terms encourage the broken edges to group into longer segments and closed boundaries, as the outlines of objects do.

Inference in the weak membrane model is hard

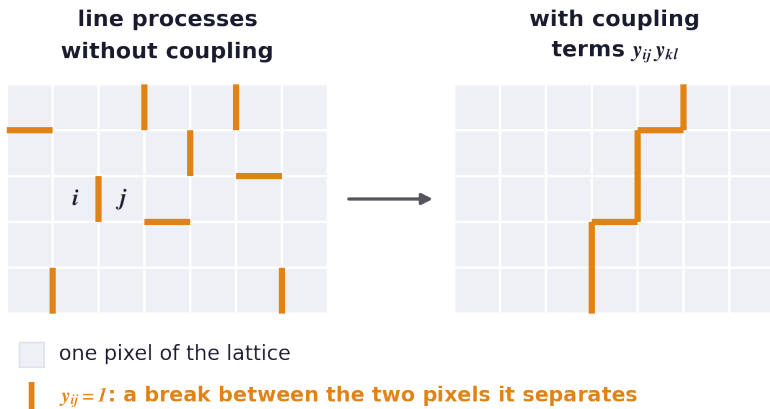


Figure 11: Line processes on a two-dimensional lattice. A grey square is a pixel. An orange segment on the side shared by two neighbouring pixels means $y_{ij} = 1$ for that pair, as between the pixels marked i and j . Without coupling the breaks are scattered fragments; with coupling terms they join end to end into one contour between two regions.

The strategy of mean field theory and variational methods

Both models ask for a search over a huge discrete set. Mean field theory and the wider family of variational methods change the question: inference becomes the minimisation of a function over simple distributions.

- ▶ Optimisation over discrete variables becomes optimisation over continuous variables, which enables steepest descent.
- ▶ Hybrid problems, with discrete and continuous variables, become continuous optimisation problems.
- ▶ Deterministic annealing introduces a temperature T , solves an easier problem at large T , and tracks the solution down to small T , where the optimisation is hard.
- ▶ MFT is a fast deterministic approximation to Markov chain Monte Carlo (MCMC), which explores the distribution by drawing random samples and is slow.
- ▶ MFT gives bounds on quantities such as the log partition function $\log Z$, which are useful for model selection: comparing two models of the same data requires their values of Z .

The strategy of mean field theory and variational methods

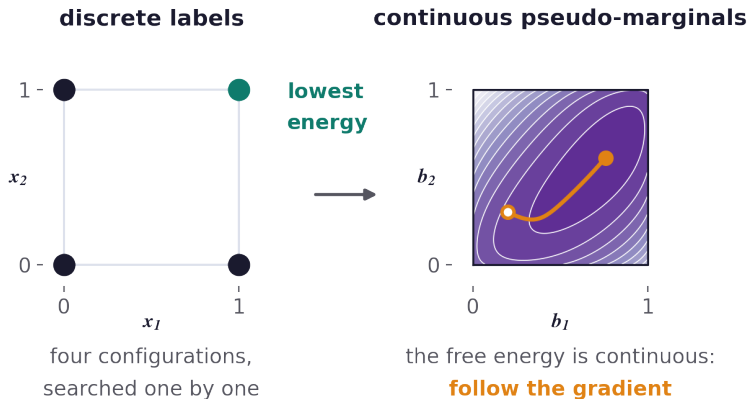


Figure 12: Two binary nodes with $\phi = (1.0, 2.2)$ and $\psi_{12} = -3.5$. The four dots are the four configurations (x_1, x_2) : $(0, 0)$ has energy 0, $(1, 0)$ has 1.0, $(0, 1)$ has 2.2, and the teal dot $(1, 1)$ has the lowest, -0.3 . The second square is the continuous version: a point (b_1, b_2) gives the probability that each node is 1, the corners are the four configurations, and the shading is the mean field free energy, darker where it is lower. The orange path is steepest descent from the open circle at $(0.2, 0.3)$ to the minimum at $(0.76, 0.61)$.

The basic idea: approximate $P(x | z)$ by a factorisable $B(x)$

Mean field theory approximates $P(x | z)$ by a simpler distribution $B^*(x)$, from which the MAP estimate and other estimators are easy to compute. Three things are specified: a class of approximating distributions, a similarity measure between a member B and the target P , and an algorithm that finds the most similar member B^* .

The class is the factorisable distributions, products of one factor per node:

$$B(x) = \prod_{i \in \mathcal{V}} b_i(x_i), \quad b_i(x_i) \geq 0, \quad \sum_{x_i} b_i(x_i) = 1.$$

Each b_i is a distribution over the values of one node, and $b = \{b_i(x_i)\}$ is the whole set. The b_i are the marginals of B . They are called pseudo-marginals because they stand in for the true marginals p_i of P , which they approximate. A factorisable distribution treats the nodes as independent, so it is described by one small table per node in place of one huge table for all of them.

The basic idea: approximate $P(\mathbf{x} \mid \mathbf{z})$ by a factorisable $B(\mathbf{x})$

The MAP estimate of $\mathbf{x} = (x_1, \dots, x_N)$ is approximated node by node,

$$\bar{x}_i = \arg \max_{x_i} b_i^*(x_i),$$

the label with the largest pseudo-marginal. The maximisation over exponentially many configurations has become N small ones.

The similarity measure is the Kullback-Leibler (KL) divergence,

$$KL(B, P) = \sum_{\mathbf{x}} B(\mathbf{x}) \log \frac{B(\mathbf{x})}{P(\mathbf{x})},$$

which is zero when $B = P$ and positive otherwise. The measure is asymmetric, and the order matters: the average is taken under B , the distribution that factorises, and that is what makes it computable.

The basic idea: approximate $P(x | z)$ by a factorisable $B(x)$

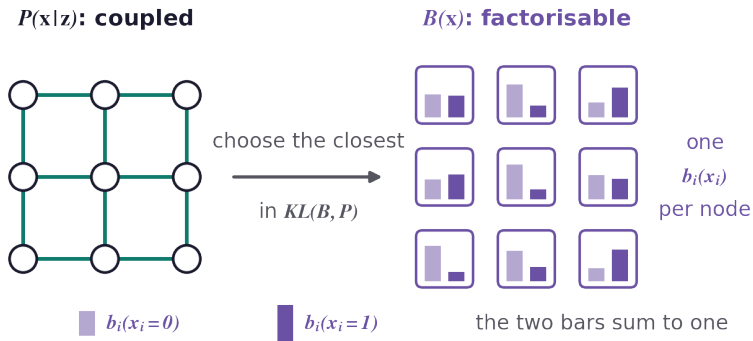


Figure 13: In $P(x | z)$ the nodes are tied together by the teal edges, so the distribution is one table over all joint configurations. In $B(x)$ the edges are gone. Each purple box is one pseudo-marginal b_i for a binary node, drawn as two bars: the pale bar is $b_i(x_i=0)$ and the dark bar is $b_i(x_i=1)$. Mean field theory chooses the heights of the bars so that B is as close to P as possible in the KL divergence.

The mean field free energy of the MRF

The quantity that mean field theory minimises is the mean field free energy, a function of the pseudo-marginals:

$$\begin{aligned}\mathcal{F}_{\text{MFT}}(\mathbf{b}) = & \underbrace{\sum_{ij \in \mathcal{E}} \sum_{x_i, x_j} b_i(x_i) b_j(x_j) \psi_{ij}(x_i, x_j)}_{\text{expected pairwise potentials}} \\ & + \underbrace{\sum_{i \in \mathcal{V}} \sum_{x_i} b_i(x_i) \phi_i(x_i, \mathbf{z})}_{\text{expected unary potentials}} + \underbrace{\sum_{i \in \mathcal{V}} \sum_{x_i} b_i(x_i) \log b_i(x_i)}_{\text{negative entropy}}.\end{aligned}$$

The first two terms are the expectation of the energy $E(\mathbf{x}, \mathbf{z})$ with respect to $B(\mathbf{x})$. They are low when the pseudo-marginals put their weight on low-energy labels. The third term is the negative entropy of $B(\mathbf{x})$, a sum over the nodes. The entropy measures how spread out a distribution is, so the third term is low when the pseudo-marginals are uncertain. The minimum of $\mathcal{F}_{\text{MFT}}$ balances the two: confident where the energy speaks clearly, uncertain elsewhere.

The expected energy of a factorisable distribution

The three terms follow from the product form of B . The expectation of one pairwise potential is

$$\sum_{\mathbf{x}} B(\mathbf{x}) \psi_{ij}(x_i, x_j) = \sum_{x_i, x_j} b_i(x_i) b_j(x_j) \psi_{ij}(x_i, x_j),$$

because every other factor b_k sums to one. The unary potentials work the same way with a single factor b_i . The entropy splits because $\log B(\mathbf{x}) = \sum_i \log b_i(x_i)$:

$$\sum_{\mathbf{x}} B(\mathbf{x}) \log B(\mathbf{x}) = \sum_{i \in \mathcal{V}} \sum_{x_i} b_i(x_i) \log b_i(x_i).$$

A sum over exponentially many configurations has become sums over nodes and edges, which can be computed.

The mean field free energy of the MRF

For binary labels $x_i \in \{0, 1\}$ write $b_i = b_i(x_i=1)$, so that $b_i(x_i=0) = 1 - b_i$, and one number per node describes B . The entropy term of a node is $b_i \log b_i + (1 - b_i) \log(1 - b_i)$. With the potentials $\phi_i(x_i) = \phi_i x_i$ and $\psi_{ij}(x_i, x_j) = \psi_{ij} x_i x_j$ the expected energy is

$$\sum_i \phi_i b_i + \sum_{ij \in \mathcal{E}} \psi_{ij} b_i b_j :$$

every label is replaced by its pseudo-marginal.

For M labels, $x_i \in \{1, \dots, M\}$, the entropy term of node i is $\sum_{l=1}^M b_{il} \log b_{il}$ with $b_{il} = b_i(x_i=l)$ and $\sum_{l=1}^M b_{il} = 1$.

The mean field free energy of the MRF

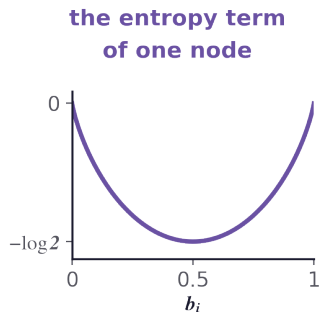
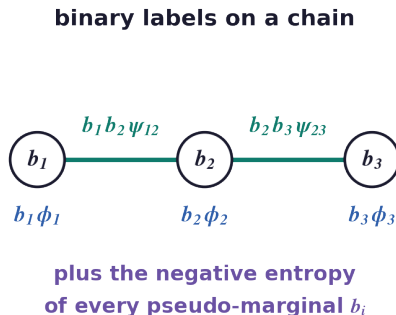


Figure 14: Where the three terms live, for binary labels on a chain of three nodes. Each node holds one number b_i . An edge contributes the teal product $b_i b_j \psi_{ij}$ and a node the blue product $b_i \phi_i$. The plot is the entropy term of one node, $b_i \log b_i + (1 - b_i) \log(1 - b_i)$: it is zero at $b_i = 0$ and $b_i = 1$, reaches its minimum $-\log 2$ at $b_i = 1/2$, is convex, which means that it curves upwards everywhere, and becomes infinitely steep at the two ends.

Minimising the KL divergence minimises the free energy

The free energy comes from the KL divergence in two steps. Split the logarithm of the ratio,

$$KL(B, P) = \sum_x B(x) \log B(x) - \sum_x B(x) \log P(x),$$

and substitute the Gibbs distribution, $\log P(x) = -E(x) - \log Z$. Since $\sum_x B(x) = 1$,

$$KL(B, P) = \sum_x B(x) E(x) + \sum_x B(x) \log B(x) + \log Z = \mathcal{F}_{\text{MFT}}(B) + \log Z.$$

Substituting the factorisable $B(x)$ turns the first two sums into the three terms of $\mathcal{F}_{\text{MFT}}(b)$. The term $\log Z$ is independent of B , so minimising the free energy gives the best factorisable approximation to $P(x)$, and the intractable Z never has to be computed.

The free energy of the segmentation lecture was the whole KL divergence; $\mathcal{F}_{\text{MFT}}$ here omits the constant $\log Z$. The minimiser is the same.

Minimising the KL divergence minimises the free energy

The same identity bounds the partition function. The KL divergence is never negative, so

$$KL(B, P) \geq 0 \implies \log Z \geq -\min_B \mathcal{F}_{\text{MFT}}(B).$$

Minus the minimum of the free energy is a lower bound on $\log Z$, and the gap is the KL divergence of the best factorisable B from P . For the chain of three binary nodes, $\log Z = 4.116$ by enumeration and the minimum of the free energy is -4.092 , so the bound 4.092 misses by 0.024 . The minimising pseudo-marginals are 0.879 , 0.787 and 0.830 , close to the true marginals 0.870 , 0.773 and 0.820 .

Minimising the KL divergence minimises the free energy

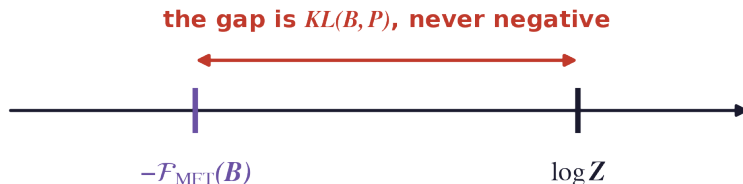


Figure 15: The two numbers on one axis. Every factorisable B gives a value $-\mathcal{F}_{\text{MFT}}(B)$ to the left of $\log Z$; the distance is $KL(B, P)$. Minimising the free energy moves the purple mark as far to the right as the factorisable family allows.

The weak membrane free energy is exact

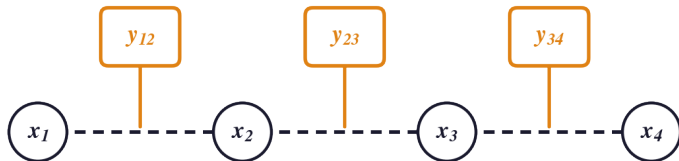
In the weak membrane the discrete variables are the line processes, so pseudo-marginals are needed for y only, with $b_{ij} = b_{ij}(y_{ij}=1)$, and x stays a continuous variable. The energy of the posterior is linear in each y_{ij} , so its expectation replaces y_{ij} by b_{ij} :

$$\begin{aligned}\mathcal{F}_{\text{MFT}}(\mathbf{b}, \mathbf{x}) = & \underbrace{\tau \sum_{i \in \mathcal{V}} (x_i - z_i)^2}_{\text{data}} + \underbrace{A \sum_{ij \in \mathcal{E}} (1 - b_{ij})(x_i - x_j)^2}_{\text{smoothness}} + \underbrace{B \sum_{ij \in \mathcal{E}} b_{ij}}_{\text{penalty}} \\ & + \underbrace{\sum_{ij \in \mathcal{E}} \{b_{ij} \log b_{ij} + (1 - b_{ij}) \log(1 - b_{ij})\}}_{\text{negative entropy of the line processes}}.\end{aligned}$$

The smoothness of an edge is weighted by $1 - b_{ij}$, the probability that the edge is intact. Given x , each y_{ij} appears in one term of the energy, $y_{ij}\{B - A(x_i - x_j)^2\}$, so the posterior $P(y | x, z)$ is a product over the edges. Given x the line processes are independent, and the factorised $B(y)$ is exact.

The weak membrane free energy is exact

given $\mathbf{x}$, each y_{ij} depends on x_i and x_j only



the line processes are independent given $\mathbf{x}$:

one pseudo-marginal $b_{ij} = b_{ij}(y_{ij} = 1)$ for every edge

Figure 16: A one-dimensional weak membrane with four pixels. The circles are the outputs x_1 to x_4 and each orange box is the line process of the dashed edge below it. With $\mathbf{x}$ held fixed, a line process touches its own two pixels only, so knowing y_{12} says nothing more about y_{23} . One number b_{ij} per edge describes the posterior of the line processes completely.

EM treats the line processes as nuisance variables

Often only the smoothed image x is wanted. The line processes are then nuisance variables: they are needed to write the model down, and they are summed out of the answer. Expectation-maximisation (EM) estimates x from

$$P(x|z) = \sum_y P(x, y|z).$$

Neal and Hinton (1998) showed that EM minimises a free energy,

$$\mathcal{F}_{\text{EM}}(B, x) = - \sum_y B(y) \log P(x, y|z) + \sum_y B(y) \log B(y).$$

Minimising $\mathcal{F}_{\text{EM}}$ with respect to B and x alternately gives the two steps of the EM algorithm. The E-step finds B , the distribution under which expectations over the hidden y are taken. The M-step maximises the expected log posterior over x , which is the same as minimising $\mathcal{F}_{\text{EM}}$ over x . For the weak membrane, $-\log P(x, y|z)$ is the energy of the posterior plus a constant, so $\mathcal{F}_{\text{EM}}(B, x)$ equals $\mathcal{F}_{\text{MFT}}(b, x)$ plus a constant.

The EM free energy is a KL divergence minus the log posterior

Write $P(x, y | z) = P(y | x, z) P(x | z)$ inside the logarithm. Since $\sum_y B(y) = 1$,

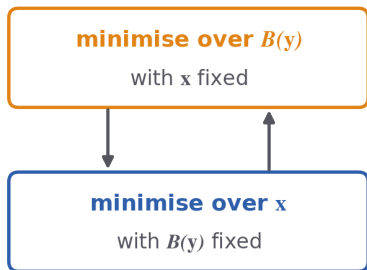
$$\mathcal{F}_{\text{EM}}(B, x) = \underbrace{\sum_y B(y) \log \frac{B(y)}{P(y | x, z)}}_{KL(B, P(y | x, z)) \geq 0} - \log P(x | z).$$

For a fixed x the best B is the posterior of the line processes, $B(y) = P(y | x, z)$. The KL divergence is then zero and the free energy equals $-\log P(x | z)$. Minimising over x as well gives the global minimum,

$$x^* = \arg \min_x \{-\log P(x | z)\}, \quad B(y) = P(y | x^*, z),$$

the most probable x with the line processes summed out. The free energy was called exact because of the step over B ; the alternation as a whole still moves downhill from its start and can stop in a local minimum.

EM treats the line processes as nuisance variables



**every half-step lowers
the free energy**

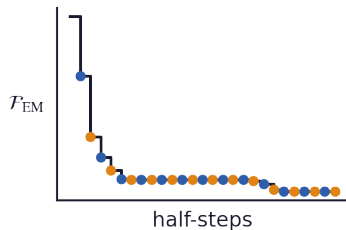


Figure 17: The two boxes are the two half-steps of the alternation, each a minimisation of the same free energy over one of its arguments with the other held fixed. The staircase is the free energy of the 60-pixel example after each half-step, without a scale: the first level is the start, a blue dot follows a minimisation over x , and an orange dot a minimisation over $B(y)$. Every step goes down.

The EM algorithm for the weak membrane

The E-step minimises $\mathcal{F}_{\text{MFT}}(\mathbf{b}, \mathbf{x})$ over each b_{ij} with $\mathbf{x}$ fixed. In the formulas a bare B is the price of a break; the distribution is always written with its argument, $B(y)$. Setting the derivative to zero, $-A(x_i - x_j)^2 + B + \log\{b_{ij}/(1 - b_{ij})\} = 0$, gives

$$\text{E-step: } b_{ij} = \frac{1}{1 + \exp\{-A(x_i - x_j)^2 + B\}}.$$

The E-step estimates the edge probabilities from the current $\mathbf{x}$: b_{ij} is the probability of a break on the edge ij of the graph, called an edge probability because a break marks an intensity edge of the image. It is $1/2$ where $A(x_i - x_j)^2 = B$, the threshold $\sqrt{B/A}$ of the edge cost, and it rises towards 1 for larger differences. With $A = 600$ and $B = 6$, a difference of 0.06 gives $b_{ij} = 0.02$ and a difference of 0.5 gives $b_{ij} = 1.00$. The hard choice between keeping and breaking an edge has become a graded one.

The EM algorithm for the weak membrane

The M-step minimises over x with b fixed:

$$\text{M-step: } x = \arg \min_x \left\{ \sum_i \tau (x_i - z_i)^2 + A \sum_{(i,j) \in \mathcal{E}} (1 - b_{ij}) (x_i - x_j)^2 \right\}.$$

The function is quadratic, so its minimum solves linear equations. Setting the derivative in x_i to zero gives

$$x_i = \frac{\tau z_i + A \sum_j (1 - b_{ij}) x_j}{\tau + A \sum_j (1 - b_{ij})},$$

the sum running over the neighbours j of i : x_i is a weighted average of the data z_i , with weight τ , and of the neighbouring outputs, with weights $A(1 - b_{ij})$. The M-step smooths z with weights $1 - b$: a probable break removes the neighbour from the average.

The EM algorithm for the weak membrane

Initialise with $x = z$ and alternate the two steps; the EM algorithm converges to a local minimum of the free energy. In the 60-pixel example the first E-step, run on the raw data, puts 22 of the 59 edge probabilities above $1/2$, because the noise alone often exceeds the threshold. After one iteration 8 remain, after three iterations 4, and after eleven iterations only the 2 true breaks, with every other b_{ij} below 0.003. The root-mean-square error of x against the noise-free signal falls from 0.059 to 0.038, 0.021 and finally 0.010.

The free energy $\mathcal{F}_{\text{MFT}}(\mathbf{b}, \mathbf{x})$ starts at 182.2 and falls within six half-steps to 34.2. It rests there while two spurious breaks at a noise spike survive, and drops to 23.5 when they disappear.

The two steps help each other. A smoother x removes the spurious breaks, and fewer breaks let the M-step average over longer stretches, which makes x smoother still.

The EM algorithm for the weak membrane

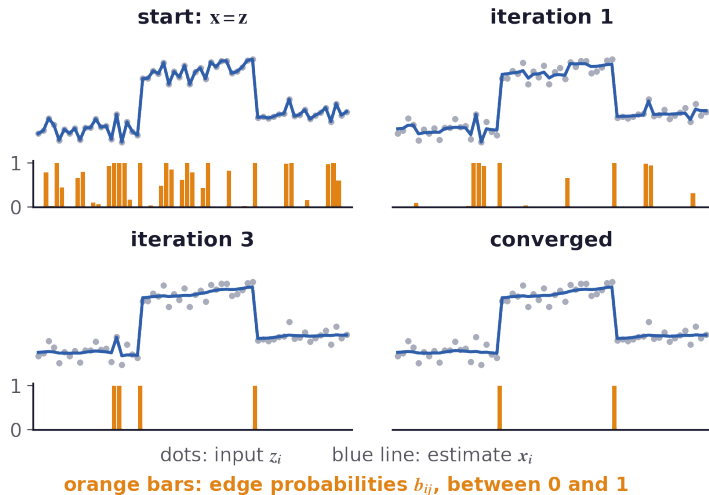


Figure 18: Four stages of the run. The upper plots show the input z_i as dots and the estimate x_i as a blue line; the lower plots show one orange bar per edge, of height b_{ij} . Bars stand everywhere at the start; two bars of height 1 are left at the end.

Interacting line processes need the mean field approximation

The modified weak membrane adds interactions between line processes to the energy, a term $C \sum y_{ij} y_{kl}$ over a set $\mathcal{E}_y$ of pairs of neighbouring line processes, for example two breaks that continue each other along a line. A negative C rewards continuing lines.

Given $\mathbf{x}$ the line processes are now coupled: their posterior no longer factorises, and the exact E-step is lost. The mean field approximation restores a factorised $B(\mathbf{y}) = \prod_{ij} b_{ij}(y_{ij})$, one pseudo-marginal for each line process. The free energy gains one term, the product of the pseudo-marginals of interacting line processes:

$$\begin{aligned}\mathcal{F}_{\text{MFT}}(\mathbf{b}, \mathbf{x}) = & \tau \sum_{i \in \mathcal{V}} (x_i - z_i)^2 + A \sum_{ij \in \mathcal{E}} (1 - b_{ij})(x_i - x_j)^2 + B \sum_{ij \in \mathcal{E}} b_{ij} \\ & + \underbrace{C \sum_{(ij) \times (kl) \in \mathcal{E}_y} b_{ij} b_{kl}}_{\text{interactions}} \\ & + \sum_{ij \in \mathcal{E}} \{ b_{ij} \log b_{ij} + (1 - b_{ij}) \log(1 - b_{ij}) \}.\end{aligned}$$

Interacting line processes need the mean field approximation

The update of one pseudo-marginal follows as in the E-step. Setting the derivative of $\mathcal{F}_{\text{MFT}}(b, x)$ in b_{ij} to zero gives

$$b_{ij} = \frac{1}{1 + \exp \left\{ -A(x_i - x_j)^2 + B + C \sum_{kl} b_{kl} \right\}},$$

where the sum runs over the line processes y_{kl} that interact with y_{ij} . Compared with the basic model, the price of a break has changed from B to $B + C \sum_{kl} b_{kl}$. With $C < 0$, probable breaks next to an edge lower the price of a break on that edge, which is how contours grow.

The update is a mean field step: each line process responds to the current pseudo-marginals of its neighbours, in place of their unknown values. The alternation with the M-step over x stays as before, and each step still lowers the free energy when the b_{ij} are updated one at a time.

Steepest descent moves downhill on the free energy

The free energy has been written down; an algorithm has to find its minimum. Mean field free energies are continuous functions of the pseudo-marginals, so gradients can be computed. Take binary labels with bilinear potentials, as in the segmentation lecture:

$$x_i \in \{0, 1\}, \quad \psi_{ij}(x_i, x_j) = \psi_{ij}x_ix_j, \quad \phi_i(x_i) = \phi_ix_i, \quad b_i = b_i(x_i=1).$$

The free energy is

$\mathcal{F}_{\text{MFT}} = \sum_{ij} \psi_{ij}b_ib_j + \sum_i \phi_ib_i + \sum_i \{b_i \log b_i + (1 - b_i) \log(1 - b_i)\}$,
and steepest descent changes each b_i at a rate equal to minus the slope of $\mathcal{F}_{\text{MFT}}$ in that direction:

$$\frac{db_i}{dt} = -\frac{\partial \mathcal{F}_{\text{MFT}}}{\partial b_i} = -\sum_j \psi_{ij}b_j - \phi_i - \log \frac{b_i}{1 - b_i},$$

where the sum runs over the neighbours j of node i . The last term is the derivative of the entropy term.

Steepest descent moves downhill on the free energy

By the chain rule the free energy decreases monotonically along the path:

$$\frac{d\mathcal{F}_{\text{MFT}}}{dt} = \sum_i \frac{\partial \mathcal{F}_{\text{MFT}}}{\partial b_i} \frac{db_i}{dt} = - \sum_i \left\{ \frac{\partial \mathcal{F}_{\text{MFT}}}{\partial b_i} \right\}^2 \leq 0.$$

The motion stops where every derivative is zero, a stationary point of $\mathcal{F}_{\text{MFT}}$. The negative entropy term keeps every b_i inside $[0, 1]$: as b_i approaches 0 the term $-\log\{b_i/(1-b_i)\}$ grows without limit and pushes b_i back up, and the mirror image happens near 1.

Variants multiply the gradient by a positive function, which keeps the decrease and improves the numerical stability. Newton-Raphson, which uses second derivatives, is another option. For the weak membrane the same recipe runs on both kinds of variables at once,

$dx_i/dt = -\partial \mathcal{F}_{\text{MFT}}(\mathbf{b}, \mathbf{x})/\partial x_i$ and $db_{ij}/dt = -\partial \mathcal{F}_{\text{MFT}}(\mathbf{b}, \mathbf{x})/\partial b_{ij}$.

Steepest descent moves downhill on the free energy

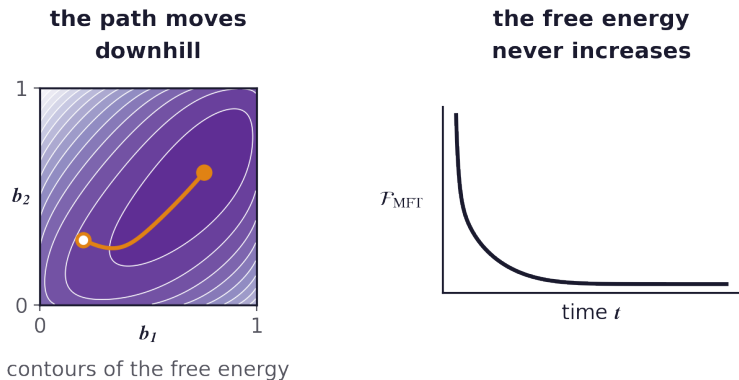


Figure 19: The two-node model with $\phi = (1.0, 2.2)$ and $\psi_{12} = -3.5$. The square is the space of pseudo-marginals (b_1, b_2) ; the shading and the white contour lines show $\mathcal{F}_{\text{MFT}}$, darker where it is lower. The orange path starts at the open circle $(0.2, 0.3)$, crosses the contours at right angles, and ends at the minimum $(0.76, 0.61)$. The second plot is $\mathcal{F}_{\text{MFT}}$ along the path as a function of time: it falls from -0.46 to -0.74 and never rises.

Hopfield analog networks: a simplified neuroscience model

One variant has a direct reading as a neural network. Multiply the gradient by the positive function $b_i(1 - b_i)$,

$$\frac{db_i}{dt} = -b_i(1 - b_i) \frac{\partial \mathcal{F}_{\text{MFT}}}{\partial b_i},$$

and change variables to $u_i = \log\{b_i/(1 - b_i)\}$, whose inverse is the sigmoid $b_i = 1/(1 + \exp\{-u_i\})$. Since $db_i/du_i = b_i(1 - b_i)$, the factor cancels, and substituting the gradient of the binary free energy gives

$$\frac{du_i}{dt} = -\frac{\partial \mathcal{F}_{\text{MFT}}}{\partial b_i} = -u_i - \sum_j \psi_{ij} b_j - \phi_i.$$

This is the dynamics of a Hopfield analog network (Hopfield, 1984). The word analog says that the output of a unit is a graded value between 0 and 1, in contrast with the two-state units of the earlier Hopfield network.

Hopfield analog networks: a simplified neuroscience model

Each unit has an internal state u_i , like the voltage of a neuron, and an output b_i between 0 and 1, like a firing rate, the number of spikes a neuron sends per second. The state integrates the weighted outputs of the other units and an external input $-\phi_i$, and the term $-u_i$ makes it decay: a leaky integrator. The weight from unit j to unit i is $-\psi_{ij}$, so a negative ψ_{ij} is an excitatory connection and a positive ψ_{ij} an inhibitory one.

At rest $u_i = -\sum_j \psi_{ij} b_j - \phi_i$, which is the mean field condition $\partial \mathcal{F}_{\text{MFT}} / \partial b_i = 0$. The network settles into a minimum of the free energy, and its outputs at rest are the pseudo-marginals. The mean field equations can therefore be read as a simplified model of a network of neurons.

Hopfield analog networks: a simplified neuroscience model

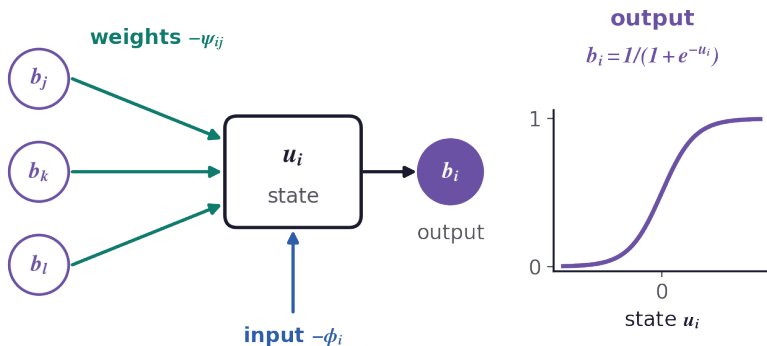


Figure 20: One unit of the network. The open purple circles are the outputs b_j , b_k , b_l of three other units; they arrive through connections of weight $-\psi_{ij}$, drawn in teal. The blue arrow is the external input $-\phi_i$. The box accumulates these into the state u_i , and the filled circle is the output b_i . The plot is the sigmoid that turns the state into the output: near 0 for a very negative state, 1/2 at $u_i = 0$, near 1 for a large state.

Discrete iterative algorithms are designed to lower the energy

On a digital computer the derivative has to be approximated by a step, $db_i/dt \approx \{b_i(t + \Delta) - b_i(t)\}/\Delta$, which gives the update $b_i(t + \Delta) = b_i(t) - \Delta \partial \mathcal{F}_{\text{MFT}} / \partial b_i$. Choosing the time step Δ is difficult: too large is unstable, too small is slow.

Discrete iterative algorithms (DIA) are designed to decrease the energy at each iteration, with no time step to choose. A DIA can make large changes of state in one iteration. Steepest descent with a small step follows every bend of the surface, which is called hugging the energy surface; a DIA jumps across.

Historically such algorithms came from fixed-point conditions, equations of the form $b = g(b)$ that the solution satisfies and that were simply iterated, and their convergence was not always guaranteed. Two strategies give algorithms with a guarantee: variational bounding and the concave-convex procedure (CCCP).

Discrete iterative algorithms are designed to lower the energy

The three behaviours can be compared on the two-node free energy, from the start (0.2, 0.3). With a time step of 0.03 steepest descent needs 136 steps to come within 0.01 of the minimum. With a time step of 0.27 it overshoots, the free energy rises, and after 6 steps the state has left the unit square.

The DIA of the example sets b_1 to the value that minimises the free energy with b_2 held fixed, then b_2 with b_1 fixed, and repeats. It arrives in 16 such updates, and the free energy falls at every one.

Discrete iterative algorithms are designed to lower the energy

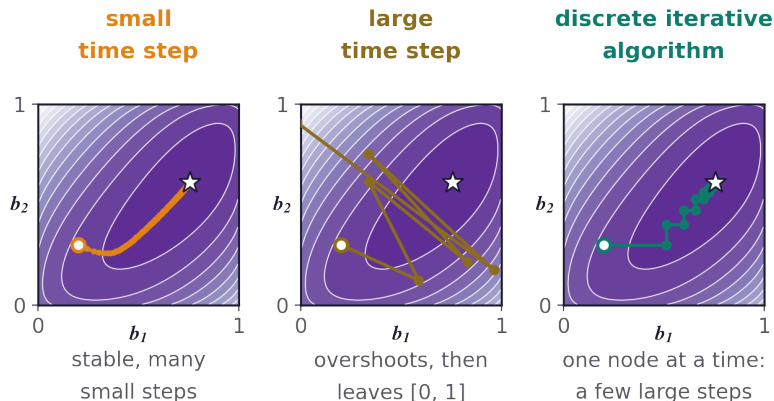


Figure 21: Three runs on the same free energy, from the same open circle; the star is the minimum. With a small time step the dots, one per step, merge into a line. With a large time step the iterates jump from side to side with growing swings and leave the square. The discrete iterative algorithm changes one coordinate at a time, so its path is a staircase of horizontal and vertical moves, each one to the lowest point along that line.

Variational bounding and the concave-convex procedure

The two strategies are best understood against steepest descent. Let the function to be minimised depend on two variables, and draw its contour lines, like the lines of equal height on a map. Steepest descent moves at right angles to the contour it stands on, the direction of the negative gradient, so its successive states, labelled x_1 , x_2 , x_3 in the figure, close in on the centre of the ellipses, where the function is lowest.

For a function of one variable with several valleys, every start inside the attraction basin of a valley slides to the bottom of that valley, even when another valley is deeper: steepest descent finds local minima. Variational bounding and CCCP keep this limitation. What they remove is the time step: each iteration jumps straight to the bottom of a curve that is known to lie above the function.

Variational bounding and the concave-convex procedure

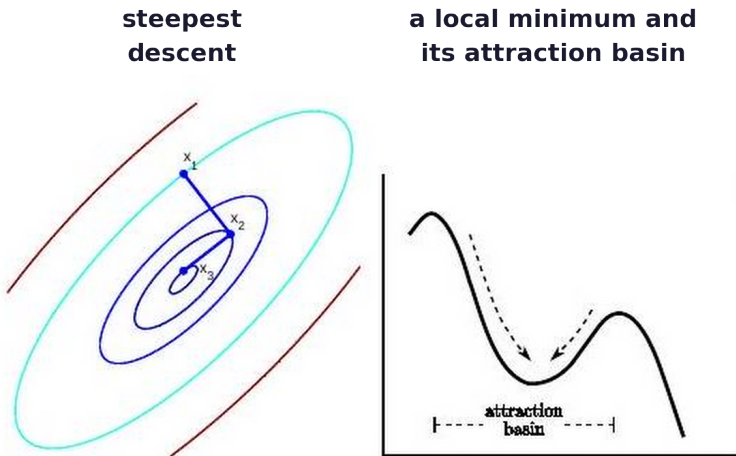


Figure 22: Steepest descent (Yuille, 2011). The closed curves are contour lines and the blue zigzag joins the states x_1 , x_2 , x_3 . The dashed arrows show descent into a local minimum, and the bracket marks its attraction basin.

Variational bounding and the concave-convex procedure

Variational bounding minimises a cost function $E(x)$ through a sequence of easier problems. At iteration t the state is x^t . Construct a bounding function $E_b(x : x^t)$ that touches E at the current state and lies above E everywhere else:

$$E_b(x^t : x^t) = E(x^t), \quad E(x) \leq E_b(x : x^t) \text{ for every } x.$$

Choose the next state x^{t+1} to lower the bound, for example at its minimum. Then

$$E(x^{t+1}) \leq E_b(x^{t+1} : x^t) \leq E_b(x^t : x^t) = E(x^t) :$$

the bound lies above E , the new state lowers the bound, and the bound touches E at the old state. The cost never increases, whatever the size of the move. The method is useful when the bounding functions can be minimised in practice.

Variational bounding and the concave-convex procedure

CCCP is the special case in which the cost is split into a convex and a concave part (Yuille and Rangarajan, 2003), $E(x) = E_{\text{vex}}(x) + E_{\text{cave}}(x)$. A convex function curves upwards and lies above all its tangents; a concave function curves downwards and lies below all its tangents. Keep the convex part and replace the concave part by its tangent at x^t :

$$E_b(x : x^t) = E_{\text{vex}}(x) + E_{\text{cave}}(x^t) + (x - x^t) \cdot \frac{\partial E_{\text{cave}}}{\partial x}(x^t).$$

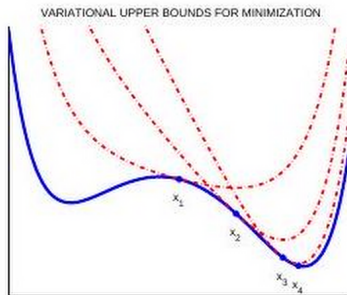
The tangent lies above the concave part and touches it at x^t , so E_b is a bounding function. It is convex, and its minimum is where its gradient vanishes:

$$\frac{\partial E_{\text{vex}}}{\partial x}(x^{t+1}) = -\frac{\partial E_{\text{cave}}}{\partial x}(x^t).$$

For free energies the entropy term is convex, which makes the procedure a natural fit.

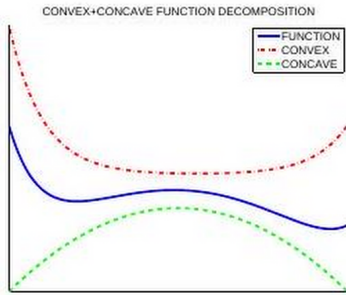
Variational bounding and the concave-convex procedure

variational upper bounds



blue: the function
red: the bounds

convex plus concave decomposition



red: convex part
green: concave part

Figure 23: Each red bound touches the blue function at the current state, x_1 , x_2 , x_3 , and its lowest point is the next state. The blue function of the second plot is the red convex curve plus the green concave curve (Yuille, 2011).

The update equation of the discrete iterative algorithm

The discrete iterative algorithm for the MRF updates the pseudo-marginal of one node from those of its neighbours:

$$b_i^{t+1}(x_i) = \frac{\exp \left\{ - \sum_j \sum_{x_j} \psi_{ij}(x_i, x_j) b_j^t(x_j) - \phi_i(x_i) \right\}}{\sum_{x'_i} \exp \left\{ - \sum_j \sum_{x_j} \psi_{ij}(x'_i, x_j) b_j^t(x_j) - \phi_i(x'_i) \right\}}.$$

The exponent is minus the expected cost of label x_i at node i : its unary potential, plus its pairwise potentials averaged over the current beliefs about the neighbours. The denominator sums the numerator over all labels x'_i and enforces $\sum_{x_i} b_i(x_i) = 1$. Cheap labels receive large pseudo-marginals.

The update equation of the discrete iterative algorithm

With the other pseudo-marginals fixed, this update minimises the free energy with respect to node i . As a function of $b_i(\cdot)$ alone, $\mathcal{F}_{\text{MFT}}$ is convex: the energy terms are linear in it and the entropy term is convex. The constraint $\sum_{x_i} b_i(x_i) = 1$ is handled by a Lagrange multiplier λ : add λ times the constraint and set the derivative to zero,

$$\sum_j \sum_{x_j} \psi_{ij}(x_i, x_j) b_j(x_j) + \phi_i(x_i) + \log b_i(x_i) + 1 + \lambda = 0.$$

Solving for $b_i(x_i)$ gives the exponential of minus the expected cost, and λ becomes the normalising denominator.

For binary labels with bilinear potentials the update is a sigmoid,

$$b_i^{t+1} = \sigma\left(-\phi_i - \sum_j \psi_{ij} b_j^t\right), \quad \sigma(u) = \frac{1}{1 + e^{-u}},$$

the update of the segmentation lecture and the resting state of the Hopfield unit.

The update equation of the discrete iterative algorithm

The order of the updates matters. One node at a time, each update is an exact minimisation over that node, so the free energy never increases. With all nodes updated in parallel from the old values the guarantee is lost, because each node answers to neighbours that are moving at the same moment.

Two binary nodes with $E(x) = 10x_1x_2 - 5x_1 - 5x_2$ show the difference. Each node alone prefers 1, and the pair is heavily penalised for being 1 together. From $b = (0.1, 0.1)$ the free energy is -1.55 . A parallel update sends both nodes to 0.98 and the free energy rises to -0.36 ; a second parallel update sends both to 0.01. Updating node 1 and then node 2 gives $(0.98, 0.01)$ and the free energy falls to -5.01 .

On a lattice a checkerboard schedule has the guarantee and most of the speed. Pixels of one colour share no edge, so updating all the black pixels together is the same as updating them one at a time, and likewise for the white pixels.

Parallel updates and the concave-convex procedure

CCCP does give a parallel update with a guarantee, for a modified free energy. In $\mathcal{F}_{\text{MFT}}$ the entropy term is convex. The energy term $\sum_{ij} \psi_{ij} b_i b_j + \sum_i \phi_i b_i$ has no terms in b_i^2 . For two nodes it is $\psi_{12} b_1 b_2$ plus linear terms, which curves upwards along $b_1 = b_2$ when $\psi_{12} > 0$ and along $b_1 = -b_2$ when $\psi_{12} < 0$. Unless every ψ_{ij} is zero, the energy term is never concave. The remedy of the chapter (Yuille, 2011) is to rewrite the binary energy as

$$E(\mathbf{x}) = \sum_{ij} \psi_{ij} x_i x_j - \alpha \sum_i x_i^2 + \sum_i \phi_i x_i + \alpha \sum_i x_i,$$

which is the same function on binary labels because $x_i^2 = x_i$. Replacing every x_i by b_i now gives an energy term that is concave once α is large enough: at least half the largest eigenvalue of the symmetric matrix that has ψ_{ij} in position ij and zeros on its diagonal.

Parallel updates and the concave-convex procedure

With the convex entropy term and the concave energy term, the CCCP condition becomes

$$b_i^{t+1} = \sigma\left(-\phi_i - \alpha - \sum_j \psi_{ij} b_j^t + 2\alpha b_i^t\right),$$

applied to all nodes in parallel. The extra term $2\alpha b_i^t$ pulls each node towards its old value, a damping that keeps the parallel step safe.

The function that decreases is $\mathcal{F}_{\text{MFT}} + \alpha \sum_i b_i(1 - b_i)$. It agrees with $\mathcal{F}_{\text{MFT}}$ wherever the pseudo-marginals are 0 or 1, and it is larger in between, so it favours decided pseudo-marginals. The guarantee therefore belongs to this update of this function. The undamped update with $\alpha = 0$, run in parallel, has no such guarantee.

The update equation of the discrete iterative algorithm

Example: binary labels from a noisy input z . A scene of 48×48 pixels holds a disk and a rectangle with label 1 on a ground with label 0, and every pixel of the input is flipped with probability 0.22. The unary potential is the cost of the observed z_i under each label, and the pairwise potential charges 1.2 for every pair of neighbours with different labels. The pseudo-marginals start from the unary potentials alone. One iteration updates the black pixels of a checkerboard from their neighbours and then the white ones; the free energy falls at every half-sweep. The fraction of wrong labels, with each pixel given its more probable label, is 23% at the start, 10% after one iteration, 3% after three and 2% after ten.

The update equation of the discrete iterative algorithm

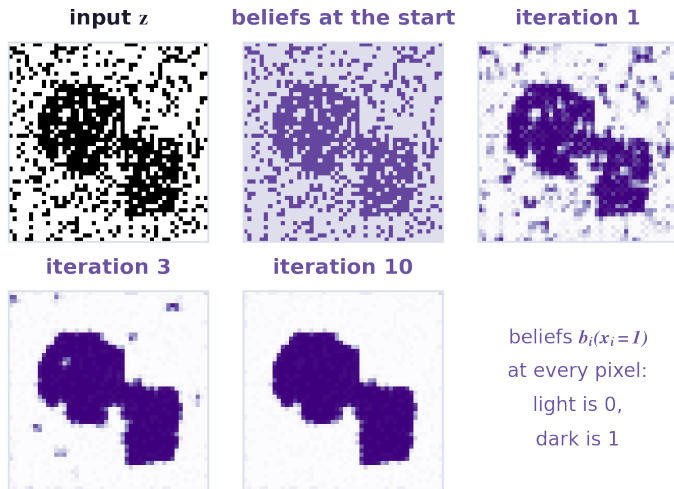


Figure 24: The input z , black for $z_i = 1$ and white for $z_i = 0$, and the pseudo-marginal $b_i(x_i=1)$ at every pixel, from white for 0 to dark purple for 1. Each iteration lets a pixel listen to its four neighbours: isolated flips fade and the shapes fill in.

A temperature parameter T defines a family of distributions

The algorithms so far find a local minimum of the free energy. A temperature parameter T gives a way to reach better minima. Define a family of distributions proportional to $\{P(x)\}^{1/T}$, which are again Gibbs distributions:

$$P(x; T) = \frac{1}{Z(T)} \exp\{-E(x)/T\}.$$

$T = 1$ is the original distribution. As T goes to 0, energy differences are magnified and the distribution peaks at the minimum of $E(x)$, the MAP estimate. As T grows, energy differences are flattened and the distribution tends to the uniform distribution. The name comes from statistical physics, where a hot system visits all its states and a cold one settles into its state of lowest energy.

A temperature parameter T defines a family of distributions

The free energy of $P(x; T)$ multiplies the entropy term by T :

$$\mathcal{F}_{\text{MFT}}(B) = \sum_x B(x)E(x) + \underbrace{T \sum_x B(x) \log B(x)}_{\text{entropy term multiplied by } T}$$

Repeating the derivation from the KL divergence with E/T in place of E shows that this is $T\{KL(B, P(\cdot; T)) - \log Z(T)\}$. Dividing by T gives the form of the segmentation lecture, E/T plus negative entropy, up to the constant $\log Z(T)$: the same minimiser.

At high temperature the entropy term dominates and the minimiser is undecided; at low temperature the expected energy dominates and the minimiser commits. For binary labels the update becomes

$$b_i = \sigma\left(-\frac{1}{T}\left\{\phi_i + \sum_j \psi_{ij}b_j\right\}\right) :$$

the temperature sets the steepness of the sigmoid.

A temperature parameter T defines a family of distributions

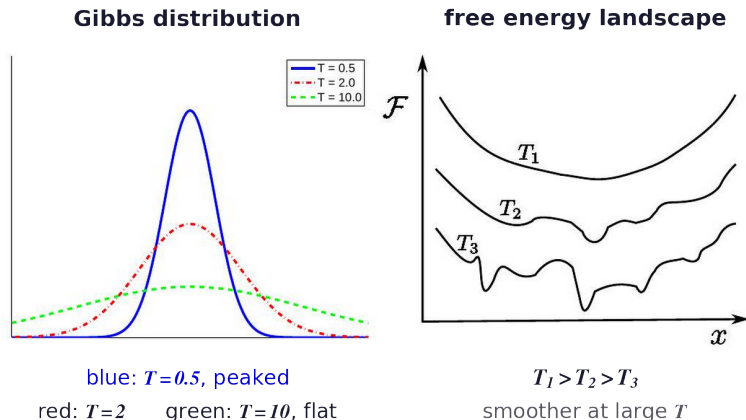


Figure 25: Two views of the temperature (Yuille, 2011). The first plot is the Gibbs distribution of one variable at three temperatures: sharply peaked at $T = 0.5$, broader at $T = 2$, almost flat at $T = 10$. The second plot sketches the free energy against the state at temperatures $T_1 > T_2 > T_3$, shifted vertically for clarity: a smooth bowl at T_1 , and more and more local minima as T falls.

Deterministic annealing tracks a minimum as T decreases

At large T the convex entropy term dominates and the free energy is convex: minimisation is easy. At small T the remaining terms dominate and the landscape is rough. Above a critical temperature T_c the free energy has an obvious solution, its single minimum, which descent reaches from any start; this is the starting point. Deterministic annealing minimises at large T , uses the result as the initial condition at a smaller T , and repeats down to small T : a continuation method, related to the continuation technique of Blake and Zisserman (1987) for the weak membrane. At sufficiently small T the global minima of the free energy approach the MAP estimates.

Two coupled binary nodes with $\phi_1 = \phi_2 = 3$ and $\psi_{12} = -7$ have energies 0 for (0,0), 3 for one node on, and -1 for (1,1): each node alone prefers 0, and together they prefer 1. On the diagonal $b_1 = b_2 = b$ the free energy is $-7b^2 + 6b + 2T\{b \log b + (1 - b) \log(1 - b)\}$, and it is convex when $T \geq T_c = 7/4$. The minimum tracked from $T = 3.2$ moves from $b = 0.58$ to 0.84 at T_c , 0.95 at $T = 1.25$ and 0.99 at $T = 0.9$, the side of the MAP estimate (1,1). At $T = 0.9$ a second minimum exists near $b = 0.05$, where a descent started near (0,0) would stop.

Deterministic annealing tracks a minimum as T decreases

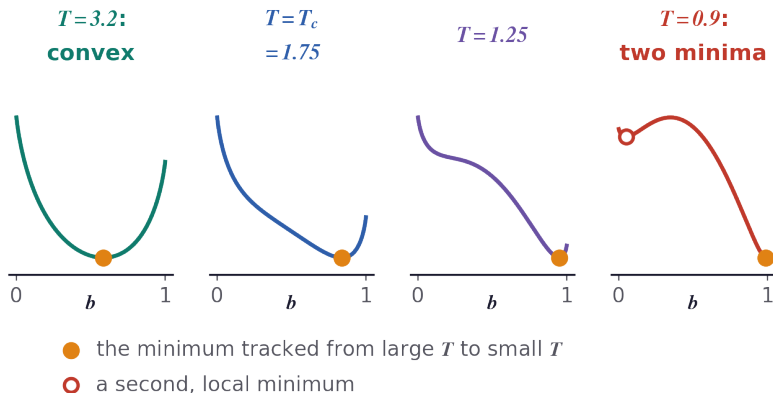


Figure 26: The free energy of the two coupled nodes along the diagonal $b_1 = b_2 = b$, at four temperatures, each panel with its own vertical scale. The orange dot is the minimum found by descent started from the orange dot of the panel before it, and the first descent starts from $b = 1/2$. The curve is a single bowl at large T , tilts as T falls, and at small T grows a second, local minimum near $b = 0$, shown as an open circle. The tracked minimum is already near $b = 1$ and stays there.

Deterministic annealing tracks a minimum as T decreases

Deterministic annealing was motivated by simulated annealing, which draws random samples from $P(x; T)$ by MCMC while T is gradually reduced. Samples at $T = 0$ sit at the global minimum of $E(x)$. Geman and Geman (1984) used simulated annealing for the weak membrane and proved that it converges to the global minimum, for a cooling schedule so slow that it is impractical; faster heuristic schedules are used in practice. Deterministic annealing has no guarantee of the global minimum. It replaces the random samples by their mean field averages and is much faster.

Stereo: the correspondence problem

The stereo MRF treated the disparity as a label at every pixel. The cooperative model looks at the same task from the side of matching, and its mean field equations have the form of a neural circuit.

The key problem is the correspondence problem: which point in the right eye's image is the projection of the same scene point as a given point in the left eye's image? The correspondence between the inputs in the two eyes gives the disparity. Given the disparity, the depth of the points in space follows by trigonometry, for calibrated cameras, whose positions and viewing directions are known.

Julesz (1971) showed that humans see depth in random dot stereograms. Both images of a random dot stereogram are fields of random black and white dots. One region of the second image is a copy of the first shifted sideways, and the strip uncovered by the shift is filled with new dots. Each image alone has no scene structure to recognise, yet viewed one by each eye the pair shows a surface floating in depth. So humans can solve stereo without first recognising objects, and an algorithm for stereo can work from the dots themselves.

Stereo: the correspondence problem

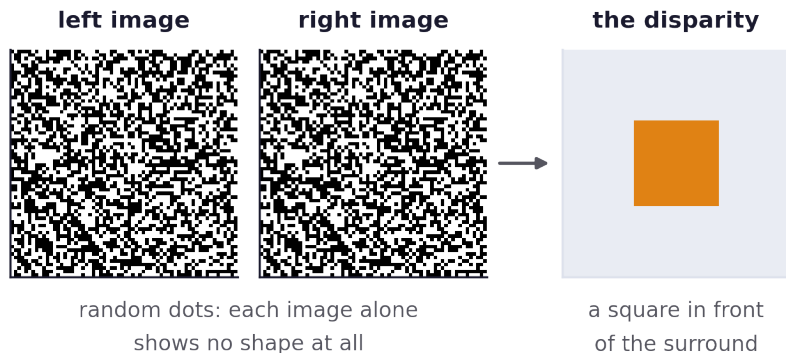


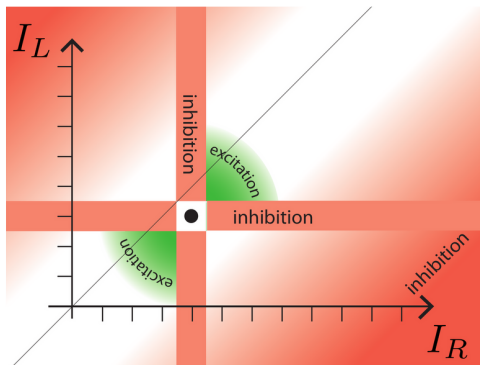
Figure 27: A random dot stereogram of 64×64 dots. The central square of 24×24 dots in the left image reappears in the right image displaced 4 dots to the left; away from the square the two images are identical. The third panel is the answer to the correspondence problem: orange where the disparity $x_R - x_L$ is -4 and grey where it is 0, a square standing in front of the surround.

Two assumptions: similar appearance and a smooth surface

Stereo algorithms address the correspondence problem with two assumptions. The first is that features in the two eyes, local descriptions of the image such as the filter responses f of the stereo MRF, are more likely to correspond if they have similar appearance. Local features alone are weak evidence: in a random dot stereogram every dot looks like every other. The second assumption is that the surface being viewed obeys weak assumptions about spatial context: it is piecewise smooth, like the weak membrane, so neighbouring points have similar disparities except at the boundaries of objects.

An early algorithm that captures the key concepts is the cooperative algorithm of Dev (1975) and of Marr and Poggio (1976). It is a Markov field model whose units are the candidate matches. Excitatory connections between matches impose the smoothness, and inhibitory connections prevent a point in one eye from having multiple matches in the other eye. The resulting algorithm combines cooperation, which implements the excitatory constraints, and competition, which deals with the inhibitory ones.

Two assumptions: similar appearance and a smooth surface



● one candidate match

■ **excitation**

neighbouring matches
at a similar disparity

■ **inhibition**

matches that share
a point in one eye,
and matches at a very
different disparity

Figure 28: The connections of one candidate match (figure: A. Yuille and D. Kersten). The horizontal axis is the position along an epipolar line of the right image I_R and the vertical axis is the position in the left image I_L , so every point of the plane is one candidate match, and the black dot is the match under consideration. Along the thin diagonal the disparity is the same as at the dot. The red bands through the dot hold the matches that use the same left point or the same right point: rivals, which are inhibited. The green lobes hold nearby matches at a similar disparity, which are excited. The red corners, far from the diagonal, hold matches at very different disparities.

A correspondence variable for every pair of points

Work on one pair of corresponding epipolar lines. The left and right images are I_L and I_R , and features are extracted at every position:

$$f(I_L) = \{f(x_L) : x_L \in \mathcal{D}_L\}, \quad f(I_R) = \{f(x_R) : x_R \in \mathcal{D}_R\},$$

where $\mathcal{D}_L$ and $\mathcal{D}_R$ are the sets of positions on the two lines. Here x_L and x_R are positions, a different use of the letter from the output x of the labelling model. For every pair of positions there is a binary correspondence variable

$$V(x_L, x_R) \in \{0, 1\}.$$

$V(x_L, x_R) = 1$ means that the two features correspond, and their positions then give the disparity $x_R - x_L$. $V(x_L, x_R) = 0$ means that the features do not match. The unknown is the whole table of these variables, and the disparity map is read off the entries that are 1. In the examples the ground has disparity 0, and a nearer surface appears further to the left in the right image, so its disparity is negative.

A correspondence variable for every pair of points

Every point is encouraged to match one other point; unmatched points and multiple matches are allowed and pay a penalty. A point can be truly unmatched when it is visible to one eye only, at the edge of a nearer surface. The same formulation applies to other matching problems, for example motion between two frames of a sequence, and even to the travelling salesman problem.

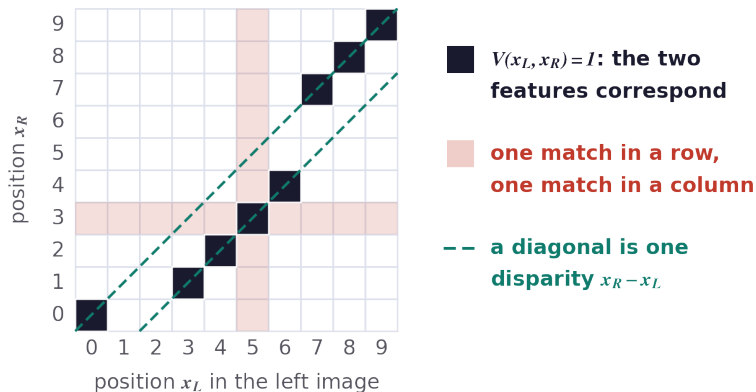
The correspondence variables of one epipolar line form a table. A column is one position x_L , a row is one position x_R , both numbered from 0, and a black cell is a match that is switched on, $V(x_L, x_R) = 1$. The connection diagram had the left image on its vertical axis; the table has it on the horizontal one. Uniqueness means at most one black cell in every column and in every row. All the cells with the same disparity $x_R - x_L$ lie on one diagonal, so a surface of constant disparity is a run of black cells along a diagonal.

A correspondence variable for every pair of points

In the example the scene is a ground at disparity 0, the main diagonal of the table, with a nearer surface at disparity -2 , the diagonal below it. The nearer surface occupies the left positions 3 to 6 and appears in the right image at positions 1 to 4.

The left positions 1 and 2 have no black cell: the nearer surface hides that part of the ground from the right eye. The rows 5 and 6 are empty: there the right eye sees ground that the left eye cannot. These are the unmatched points that the penalty has to tolerate at every depth edge.

A correspondence variable for every pair of points



constant disparity: the matches stay on one diagonal;
at a depth edge some points stay unmatched

Figure 29: The correspondence variables for ten positions in each image. The red column and row mark the cells that compete with the match at their crossing; the dashed diagonals are the disparities 0 and -2 .

The energy: feature match, uniqueness and smooth disparity

The model is a Gibbs distribution over the correspondence variables,

$$P(V | f(I_L), f(I_R)) = \frac{1}{Z} \exp \{ - E(V; f(I_L), f(I_R)) / T \},$$

with three kinds of terms in the energy:

$$\begin{aligned} E = & \underbrace{\sum_{x_L, x_R} V(x_L, x_R) M(f(x_L), f(x_R))}_{\text{matches between similar features}} \\ & + \underbrace{A \sum_{x_L} \left(\sum_{x_R} V(x_L, x_R) - 1 \right)^2 + A \sum_{x_R} \left(\sum_{x_L} V(x_L, x_R) - 1 \right)^2}_{\text{every point matched once}} \\ & + \underbrace{C \sum_{x_L, x_R} \sum_{\substack{y_L \in N(x_L) \\ y_R \in N(x_R)}} V(x_L, x_R) V(y_L, y_R) \{ (x_R - x_L) - (y_R - y_L) \}^2}_{\text{neighbouring points have similar disparities}}. \end{aligned}$$

The energy: feature match, uniqueness and smooth disparity

The first term is the evidence. M takes small values if the two features are similar and large values if they are different, so a match between similar features is cheap. The constants A and C are new and have no relation to those of the weak membrane.

The second pair of terms is the uniqueness constraint. For a left point x_L the bracket counts its matches and subtracts one. The square is 0 for exactly one match, 1 for an unmatched point and 1 for two matches. The same holds for every right point.

The third term is the smoothness prior. $N(\cdot)$ is a spatial neighbourhood, so (y_L, y_R) is a match between points next to x_L and next to x_R . Two such matches that are both on pay C times the squared difference of their disparities, once for each order of the pair. The term only charges disagreement; matches that agree are favoured relative to those that disagree, and in this sense they support each other.

T is the temperature, with default value $T = 1$. Two-dimensional images are handled one epipolar line at a time. The simplest minimisation flips one variable at a time and keeps the flip if the energy decreases.

Mean field theory gives a neural circuit model

Mean field theory replaces each binary $V(x_L, x_R)$ by a continuous $q(x_L, x_R) \in [0, 1]$, the pseudo-marginal of $V(x_L, x_R) = 1$, which plays the part of b . The free energy $\mathcal{F}$ is the expected energy plus T times the negative entropy of the q . With the state $u = T \log\{q/(1 - q)\}$, so that $q = 1/(1 + e^{-u/T})$, the Hopfield dynamics $du/dt = -\partial\mathcal{F}/\partial q$ are

$$\begin{aligned} \frac{du(x_L, x_R)}{dt} = & \underbrace{-u(x_L, x_R)}_{\text{integration}} - \underbrace{M(f(x_L), f(x_R))}_{\text{feature match}} \\ & \underbrace{-2A\left(\sum_{y_R \neq x_R} q(x_L, y_R) - \frac{1}{2}\right) - 2A\left(\sum_{y_L \neq x_L} q(y_L, x_R) - \frac{1}{2}\right)}_{\text{inhibition between competing matches}} \\ & \underbrace{-2C \sum_{\substack{y_L \in N(x_L) \\ y_R \in N(x_R)}} q(y_L, y_R) \{(x_R - x_L) - (y_R - y_L)\}^2}_{\text{neighbours at a different disparity suppress}}. \end{aligned}$$

The uniqueness term of a factorisable distribution

The inhibition comes from the expectation of the uniqueness term. Take one left point and call its candidate matches $V_1, \dots, V_n$, independent under the factorisable distribution, with $V_r = 1$ with probability q_r . Angle brackets denote the expectation under that distribution. The number of matches $\sum_r V_r$ has mean $\sum_r q_r$ and variance $\sum_r q_r(1 - q_r)$, so

$$\left\langle \left(\sum_r V_r - 1 \right)^2 \right\rangle = \sum_r q_r(1 - q_r) + \left(\sum_r q_r - 1 \right)^2 = 1 - \sum_r q_r + 2 \sum_{r < s} q_r q_s.$$

The derivative with respect to one q_r is

$$-1 + 2 \sum_{s \neq r} q_s = 2 \left(\sum_{s \neq r} q_s - \frac{1}{2} \right).$$

A match is pushed down when its rivals together hold more than half a match, and pushed up when they hold less: with no rivals at all the term drives the unit on, since an unmatched point is also penalised. In the smoothness term every pair of neighbouring matches is counted in both orders, which gives the factor 2C.

Mean field theory gives a neural circuit model

Each unit $u(x_L, x_R)$ behaves like a neuron tuned to one position and one disparity: it responds most strongly to a match there. Each term of the update is a kind of input. The first is the standard integration term: left alone, the state decays. The second is the drive from the images: a good feature match has a small M and subtracts little. It can come from local neurons tuned to disparity. Take the cyclopean position $\frac{1}{2}(x_R + x_L)$, the position seen from midway between the eyes, and let $v(\bar{D})$ be the summed response there of the neurons that prefer disparity D ; then $M = \exp\{-v(D)\}$ turns a strong response into a small M .

The third and fourth terms are inhibition between competing matches, the units that share x_L or share x_R . The last term carries the smoothness. With $C > 0$ it is never positive: neighbouring matches at a different disparity suppress the unit, and those at the same disparity leave it alone. Relative to their rivals, the matches of a smooth surface are therefore favoured, which is the cooperation of the cooperative algorithm. An explicitly excitatory connection, as in the network of Marr and Poggio, corresponds to a negative energy for compatible pairs.

Cooperation and competition sharpen the matches

The network is run on one epipolar line of a random dot stereogram with 48 dots: a surface in front of a ground. The dots from 16 to 31 belong to the nearer surface, at disparity $x_R - x_L = -3$, and the others to the ground, at disparity 0. Disparities from 0 to -5 are allowed. M is the mean absolute difference of the dots in a window of five, multiplied by 6, and $A = 1$, $C = 0.35$, $T = 1$.

From the features alone every point has several candidate matches, because random dots look alike: a window of five binary dots finds accidental copies of itself. Choosing the best feature match at every point gives 14 wrong disparities out of 48. After the network settles, one match survives at almost every point and it lies on the true disparity. Four errors are left, at points 14, 15, 16 and 31, all at the two depth edges, where some dots are visible to one eye only.

Cooperation and competition sharpen the matches

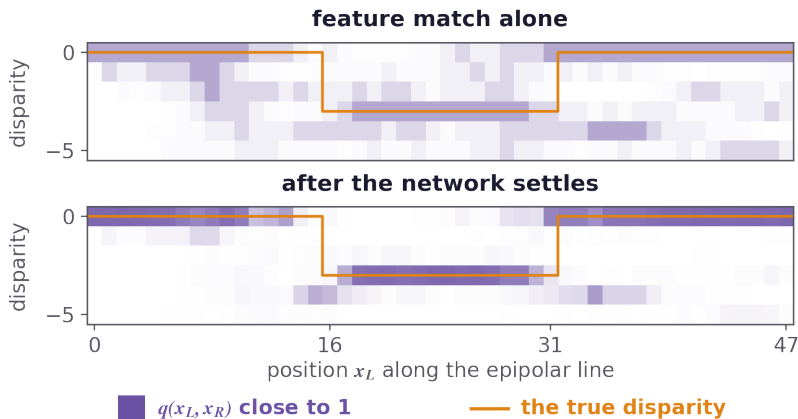


Figure 30: Each panel is a table of units: a column is a position x_L on the left epipolar line, a row is a disparity $x_R - x_L$ from 0 at the top to -5 at the bottom, and the shade of a cell is $q(x_L, x_R)$, from white for 0 to purple for 1. With the feature match alone, several cells are shaded in most columns. After the network settles, the purple has collected along the orange line, the true disparity.

Electrophysiology is consistent with the cooperative model

Is there evidence for such a circuit in the brain? Electrophysiology records the electrical activity of neurons. Samonds, Potetz and Lee (2009) recorded from many neurons at once in the primary visual cortex (V1) of macaque monkeys with an array of electrodes, while the animals viewed random dot stereograms. Many V1 neurons are tuned to disparity: the firing rate of a neuron plotted against the disparity of the stimulus, its tuning curve, has a peak at the preferred disparity of the neuron.

The recordings agree in general with the predictions of this type of algorithm. Neurons tuned to similar disparities at neighbouring positions excite each other, and neurons tuned to different disparities at the same position inhibit each other: the two kinds of connection of the cooperative model.

In the plots of the experiment the firing rates are normalised, divided by the largest rate of the neuron, and the disparity is measured as a visual angle, from -1 to 1 degree.

Electrophysiology is consistent with the cooperative model

recordings from many neurons at once

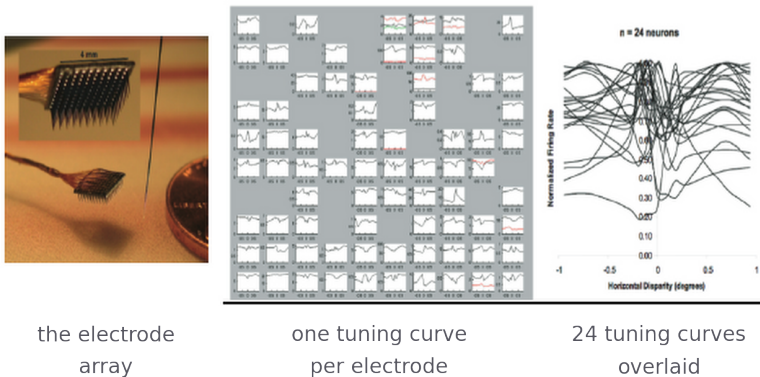


Figure 31: Recording disparity tuning from many V1 neurons at once (Samonds et al., 2009). The photograph shows the array of electrodes, 4 mm across, next to a coin. The grey panel has one small plot per electrode, laid out as on the array: the tuning curve of the neuron recorded there. The last panel overlays 24 tuning curves.

Electrophysiology is consistent with the cooperative model

A variant of the stereo model accounts for further findings (Samonds, Potetz, Tyler and Lee, 2013). The disparity tuning of the neurons sharpens over the first few hundred milliseconds after the stimulus appears, as it does in a network that settles. The model also reproduces the responses to anticorrelated stereograms, in which the left and right images have opposite contrast. Detailed biological evidence for these models remains preliminary.

Every plot of the comparison shows normalised firing rate against binocular disparity in degrees. The column headed Model compares the input to the network, in grey, with its steady state, in black. The column headed Recordings shows V1 responses in four time windows between 100 and 850 ms after stimulus onset, the later windows in darker and thicker lines. The top row is for dynamic random dot stereograms (DRDS), whose dots are redrawn many times a second: the peak narrows and the flanks fall, in the model and in the neurons. The bottom row is for anticorrelated ones (a-DRDS), where the tuning curve is inverted and deepens in the same way.

Electrophysiology is consistent with the cooperative model

tuning sharpens over time

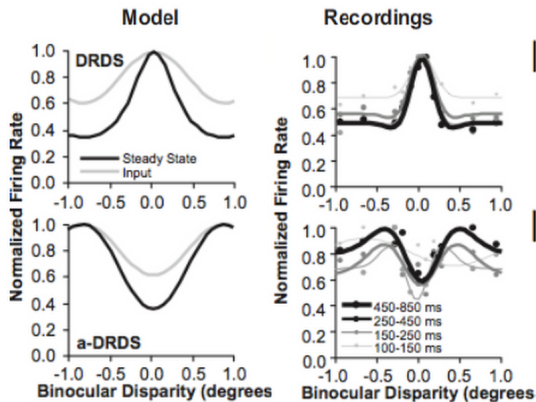


Figure 32: Disparity tuning sharpens over time, in the model and in the recordings, for correlated (DRDS) and anticorrelated (a-DRDS) random dot stereograms (Samonds et al., 2013).

Bayesian theories are said to need slow relaxation algorithms

Every algorithm so far is a relaxation algorithm: it starts from a guess and updates it many times until it settles, as steepest descent, the discrete iterative algorithms and the stereo network all do. Bayesian vision theories are often claimed to require such time-consuming relaxation, which looks biologically implausible.

Smirnakis and Yuille (1995) show that Bayesian theories can in fact be implemented by feedforward networks. A feedforward network, or multilayer perceptron, passes its input through layers of units, each unit taking a weighted sum of the layer before and applying a sigmoid. One pass gives the output.

The weights are trained by unsupervised learning with a variant of backpropagation, the algorithm that computes the gradient of a training criterion with respect to all the weights. Unsupervised means that no desired outputs are given, so no teacher is needed; the prior and the likelihood of the theory are given. Feedback loops are required during training only; after learning the network is feedforward. Multilayer perceptrons and backpropagation are chosen for convenience.

Bayesian theories are said to need slow relaxation algorithms

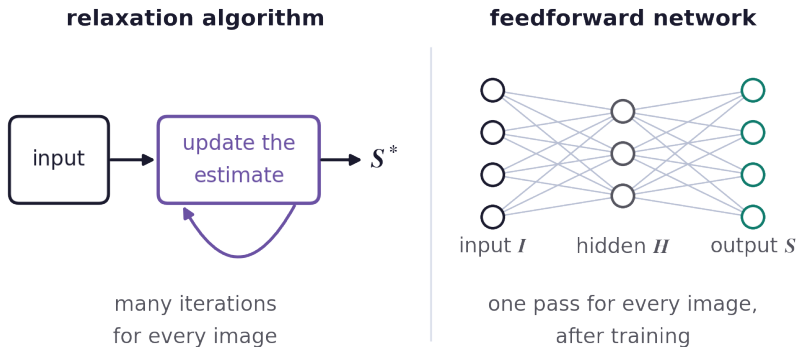


Figure 33: Two ways to compute the estimate S^* of a Bayesian theory. In a relaxation algorithm the box is visited again and again, the purple arrow feeding each estimate back in as the start of the next update, and this loop runs for every new image. In the feedforward network each circle is a unit and each grey line a weight; the input I enters on one side, passes the hidden units H once, and the output S is read off the other side.

The network is trained to maximise the posterior of its output

In a visual task, S are the world properties to extract and I is the visual input: the x and z of the earlier models. A Bayesian theory supplies a likelihood $P_I(I | S)$ and a prior $P_{\text{prior}}(S)$, and asks for the MAP estimator:

$$P_{\text{post}}(S | I) = \frac{P_I(I | S) P_{\text{prior}}(S)}{P(I)}, \quad S^*(I) = \arg \max_S P_{\text{post}}(S | I).$$

A relaxation algorithm solves this maximisation afresh for every input. The alternative is to learn the function $S^*(\cdot)$ once. Let a network with one hidden layer and weights ω compute $S = f(I; \omega)$, and choose the weights that make the outputs as probable as possible over a set Λ of representative inputs I^μ :

$$\omega^* = \arg \max_{\omega} \sum_{\mu \in \Lambda} \log P_{\text{post}}(f(I^\mu; \omega) | I^\mu).$$

The criterion scores the network's own output with the posterior of the theory.

The network is trained to maximise the posterior of its output

As the set of inputs grows, the average over the set tends to $\sum_I P(I) \log P_{\text{post}}(f(I; \omega) | I)$, so $P(I)$ never has to be known: the inputs that occur are the sample. If the class of networks contains the MAP estimator, the criterion is maximised by weights with $f(I; \omega^*) = S^*(I)$; otherwise the result is the best approximation within the class.

Training is backpropagation with $\log P_{\text{post}}$ in the place of the error function, climbing where ordinary training descends. For a Gibbs posterior, $P_{\text{post}}(S | I) = e^{-E_{\text{post}}(S, I)} / Z(I)$, the constant $\log Z(I)$ does not depend on the weights, so the network minimises the energy of its own outputs, $\sum_{\mu} E_{\text{post}}(f(I^{\mu}; \omega), I^{\mu})$. In ordinary supervised training the error at the output is the difference from a desired output. Here it is the gradient of the energy, $\partial E_{\text{post}} / \partial S$, which is propagated back through the network to change the weights. Presenting the inputs one at a time in random order makes the training noisy, which helps the weights escape poor local optima.

The network is trained to maximise the posterior of its output

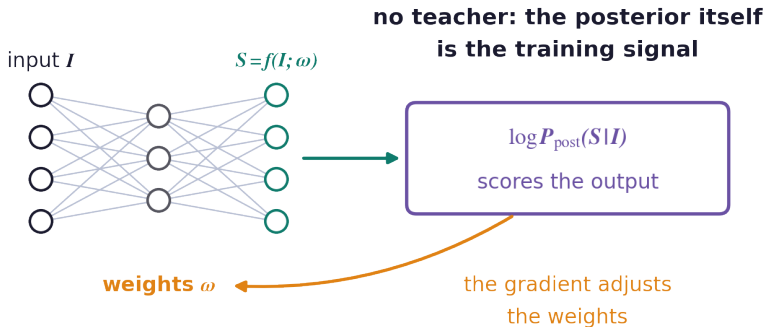


Figure 34: One step of training. An input I passes through the network and gives the output $S = f(I; \omega)$. The purple box is the theory: it returns the log posterior of that output. The orange arrow is the gradient, which adjusts the weights so that the next output scores higher.

Simulation example: image segmentation by the weak string

The theory is tested on the weak string, the weak membrane in one dimension: smooth a noisy signal I into an output S and preserve the edges. The posterior is a Gibbs distribution with energy

$$F^p(S) = \sum_{i=0}^N (S_i - I_i)^2 + \sum_{i=1}^N g^p(S_i - S_{i-1}),$$

$$g^p(t) = \begin{cases} \lambda^2 t^2 & |t| < q \\ \alpha - c (|t| - r)^2 / 2 & q \leq |t| < r \\ \alpha & |t| \geq r, \end{cases}$$

$$c = \frac{1}{2p}, \quad r^2 = \alpha \left(\frac{2}{c} + \frac{1}{\lambda^2} \right), \quad q = \frac{\alpha}{\lambda^2 r}.$$

The first sum is the data term. The function g^p charges the difference t between neighbours: a quadratic smoothness cost for small $|t|$, a constant α for large $|t|$, and a concave arc that joins the two. Here q and r are the break points of g^p , and $\alpha = 0.025$, $\lambda = 5$; the letters q , α and λ have no relation to their earlier uses.

Simulation example: image segmentation by the weak string

In the limit $p \rightarrow 0$ the arc shrinks to nothing and $g^0(t) = \min(\lambda^2 t^2, \alpha)$. This is the edge cost of the weak membrane with the line process minimised out, with $\tau = 1$, $A = \lambda^2$ and $B = \alpha$. Output differences above $\sqrt{\alpha}/\lambda = 0.032$ are cheaper to break, and an isolated jump of the input is kept as an edge once it exceeds about 0.1.

The parameter p comes from the graduated non-convexity method of Blake and Zisserman (1987). At $p = 0$ the energy has many local minima, which would trap the training. At $p = 1$ the arc is wide, from 0.003 to 0.32, and its downward curvature $c = 1/2$ is small enough for the data term to outweigh it, so $F^1(S)$ is convex in S . Graduated non-convexity starts at $p = 1$ and lowers p step by step to 0 while the solution is tracked. It is a continuation method, with p in the role that the temperature T plays in deterministic annealing.

Simulation example: image segmentation by the weak string

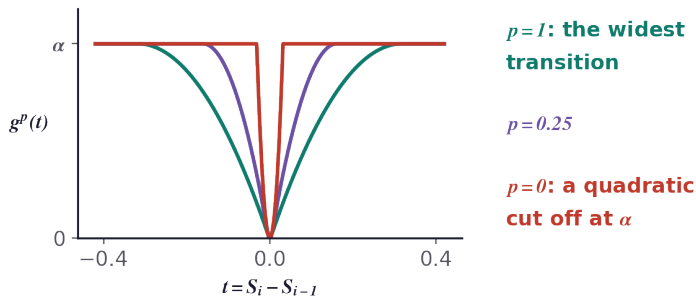


Figure 35: The smoothness cost $g^p(t)$ of one pair of neighbours against their difference t , for $p = 1$ in teal, $p = 0.25$ in purple and $p = 0$ in red. All three curves rise from 0 and level off at α . The transition is widest at $p = 1$ and sharpens as p falls; at $p = 0$ the curve is the quadratic cut off at α .

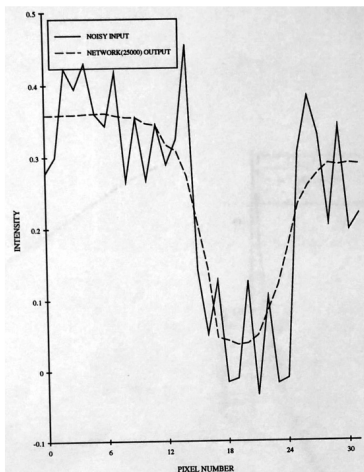
The network smooths the noise and preserves the edges

Training in one dimension used 1000 simulated patterns of length 32: piecewise constant signals with values in $[-0.5, 0.5]$ plus uniform noise in $[-0.2, 0.2]$. The network has three fully connected layers of 32 sigmoidal units, with outputs in $[-0.5, 0.5]$. The noise-free patterns are the outputs a good estimator should give; they serve to judge the result and are never used as a teaching signal.

After training, the network reproduces the noise-free patterns of its training set, reduces the noise of patterns with added noise, and gives piecewise constant outputs with the edges in the right places. The output is smooth, and the intensity edges and their locations are preserved: the behaviour that the weak string energy asks for, obtained in a single pass.

In the example from the paper the horizontal axis is the pixel number, 0 to 31, and the vertical axis the intensity. The solid line is a noisy input with a high level, a drop near pixel 14 and a rise near pixel 24. The dashed line is the output of the network after 25,000 training iterations: smooth on each of the three pieces, with two transitions centred close to the two jumps of the input.

The network smooths the noise and preserves the edges



solid line:
the noisy input

**dashed line: the output
of the network**

the noise is smoothed away;
the two intensity edges
keep their locations

horizontal axis: pixel number
vertical axis: intensity

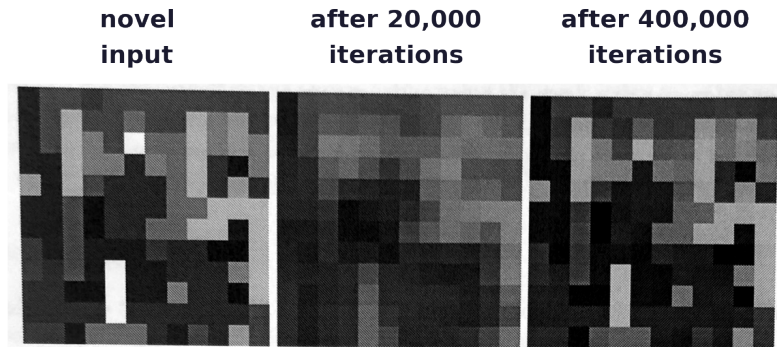
Figure 36: A noisy input and the output of the trained network (Figure 1 of Smirnakis and Yuille, 1995).

The two-dimensional network generalises to novel inputs

The two-dimensional version implements the weak membrane on simulated images of 12×12 pixels, a direct generalisation of the network for one dimension, trained on 1000 random patterns. The network memorises the training set, reduces noise, and generalises to edge detection on novel inputs, images that were never part of the training set.

The conclusion is that a Bayesian theory can be implemented in a feedforward manner, faster than by relaxation and more plausible biologically. The slow search has moved from perception into learning: it is done once, over many images, and every later image costs one pass. The priors and likelihoods that drive the learning could be fixed by evolution, so unsupervised learning of this kind might play a role in the development of the visual system.

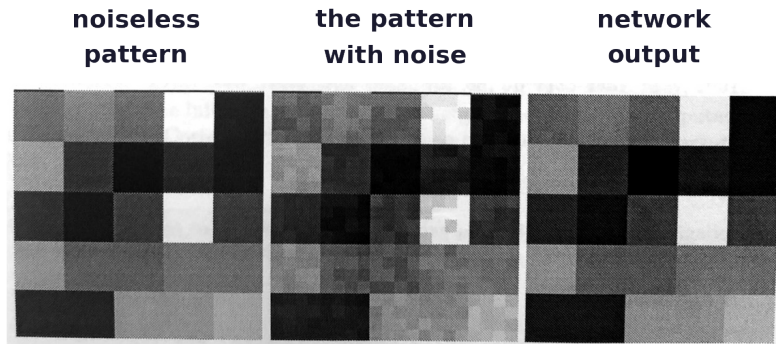
The two-dimensional network generalises to novel inputs



20,000 and 400,000 are numbers of training iterations

Figure 37: Figure 3 of Smirnakis and Yuille (1995). Each panel is an image of 12×12 pixels shown as grey squares. The first is an input that was never used in training. The second is the output after 20,000 training iterations, a heavy blur. The third is the output after 400,000: regions of nearly constant grey with sharp borders in the places of the input's borders.

The two-dimensional network generalises to novel inputs



uniform noise of amplitude 0.1 on every pixel

Figure 38: Figure 4 of Smirnakis and Yuille (1995): a piecewise constant pattern, the same pattern with uniform noise of amplitude 0.1 added at every pixel, and the output of the network for the noisy pattern. The speckle inside the blocks is gone and the borders of the blocks are unchanged, so the network has learnt more than to copy its input.

Summary

- ▶ Image labelling, binocular stereo and the weak membrane are Markov models: an energy on a graph.
- ▶ The weak membrane adds hidden line processes that switch off the smoothness at a discontinuity.
- ▶ Mean field theory approximates $P(x|z)$ by a factorisable $B(x)$ and minimises a free energy.
- ▶ The free energy bounds $\log Z$; for the weak membrane the free energy is exact and leads to an EM algorithm.
- ▶ Steepest descent, discrete iterative algorithms and CCCP minimise the free energy; annealing varies T .
- ▶ Mean field theory on the stereo energy gives a network of competing and mutually supporting matches.
- ▶ Recordings in primary visual cortex are consistent with these cooperative and competitive interactions.
- ▶ A feedforward network trained without a teacher can approximate the MAP estimate in one pass.

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